

Revealing a hidden conducting state by manipulating the intracellular domains in $K_V10.1$ exposes the coupling between two gating mechanisms

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Abstract

The *KCNH* family of potassium channels serves relevant physiological functions in both excitable and non-excitable cells, reflected in the massive consequences of mutations or pharmacological manipulation of their function. This group of channels shares structural homology with other voltage-gated K^+ channels, but the mechanisms of gating in this family show significant differences with respect to the canonical electromechanical coupling in these molecules. In particular, the large intracellular domains of *KCNH* channels play a crucial role in gating that is still only partly understood. Using *KCNH1*($K_V10.1$) as a model, we have characterized the behavior of a series of modified channels that could not be explained by the current models. With electrophysiological and biochemical methods combined with mathematical modeling, we show that the uncovering of an open state can explain the behavior of the mutants. This open state, which is not detectable in wild-type channels, appears to lack the rapid flicker block of the conventional open state. Because it is accessed from deep closed states, it elucidates intermediate gating events well ahead of channel opening in the wild type. This allowed us to study gating steps prior to opening, which, for example, explain the mechanism of gating inhibition by Ca^{2+} -Calmodulin and generate a model that describes the characteristic features of *KCNH* channels gating.

eLife assessment

This **valuable** study examines the role of the interaction between cytoplasmic N- and C-terminal domains in voltage-dependent gating of Kv10.1 channels. The authors claim to have identified a hidden open state in Kv10.1 mutant channels, thus providing a window for observing early conformational transitions associated with channel gating. The evidence supporting the major conclusions is **incomplete**, however, and additional work is required to determine the molecular mechanism underlying the observations in this study. With the experimental conditions clarified and the mechanistic interpretations addressed, this work could be significant in understanding the gating mechanisms of the KCNH family and will appeal to biophysicists interested in ion channels and physiologists interested in cancer biology.

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Introduction

Voltage-gated potassium channels constitute a large family of proteins that allow K^+ flow upon changes in the membrane potential. Their general architecture consists of a tetrameric complex with six transmembrane segments (S1-S6) in each subunit to form a central pore with four voltage sensors at the periphery. S1 to S4 segments constitute the sensor domain, while S5 and S6 segments, together with the loop linking them, line the pore. The mechanism of voltage-dependent gating is well understood for several subfamilies, namely those closely related to the *Drosophila Shaker* channel (*KCNA* and *KCNB*). In this subset of families, the linker between the sensor and pore domains (S4-S5 linker) acts as a mechanical lever transferring the movement of the voltage sensor to the gate at the bottom of S6 in a neighboring subunit through physical interaction (Barros et al. 2020). Such trans-subunit interaction is commonly denoted “domain swapping.” In other families, like the EAG family (*KCNH*) (Malak et al. 2019; Tomczak et al. 2017; Whicher and MacKinnon 2019), gating mechanics must be different because there is no domain swapping in the transmembrane regions, the S4-S5 segment is very short and does not form a helix (Whicher and MacKinnon 2016; Wang and MacKinnon 2017), and voltage-dependent gating occurs even when the S4-S5 linker is severed or removed in Kv10.1 (Lörinczi et al. 2015) or Kv11.1 (de la Peña et al. 2018). A thorough understanding of gating mechanisms in this family is also relevant in practical terms because members of the family are implicated in many pathological conditions, as is the case for Kv101 (Bauer and Schwarz 2018; Toplak et al. 2022), which makes these channels attractive therapeutic targets.

Although *KCNH* channels do not display domain-swapping in the transmembrane domains, they show extensive conserved intracellular domains where intersubunit interactions occur. The so-called eag domain in the N-terminus, formed by the PAS domain and PAS-Cap, interacts with the CNBHD (cyclic nucleotide-binding homology domain, see Fig. S1A) of the neighboring subunit, which is connected to the S6 through the C-linker. The four eag-CNBHD complexes form a ring in the intracellular side of the channel, connected to the gate via the C-linkers (Whicher and MacKinnon 2019; Whicher and MacKinnon 2016). Thus, there is domain-swapping within intracellular domains instead of the transmembrane segments (reviewed, e.g., in (Barros et al. 2020; Codding et al. 2020)). This arrangement makes the intracellular ring an excellent candidate to participate in the gating process, as it has been demonstrated for other channels, e.g. (Codding et al. 2020; James and Zagotta 2017; Nunez et al. 2020; Verkest et al. 2022). The first residues of the PASCap segment, although unresolved in the available structures of Kv10.1,

occupy in Kv11.1 a space close to the S4-S5 linker and the C-linkers, suggesting its participation in transducing the movements of the voltage sensor to the intracellular ring (Wang and MacKinnon 2017 [↗](#); Stevens-Sostre et al. 2024 [↗](#)). Indeed, the gating of *KCNH* channels is affected by manipulations of either the eag domain, the CNBHD, or the interaction between them (Stevens-Sostre et al. 2024 [↗](#); Malak et al. 2019 [↗](#); Whicher and MacKinnon 2019 [↗](#); Coddington et al. 2020 [↗](#); Coddington and Trudeau 2019 [↗](#); Dai and Zagotta 2017 [↗](#); Dai et al. 2018 [↗](#); Gianulis et al. 2013 [↗](#); Ju and Wray 2006 [↗](#); Sahoo et al. 2012 [↗](#); Terlau et al. 1997 [↗](#)). Furthermore, the interaction between the PAS domain and the C-terminus is more stable in closed than in open Kv11.1 (HERG) channels, and a single chain antibody binding to the interface between PAS domain and CNBHD can access its epitope in open but not in closed channels, strongly supporting a change in conformation of the ring during gating (Harley et al. 2021 [↗](#)). Recent cryo-EM work revealed that the position of the voltage sensor in a polarized membrane would preclude the opening of the gate (Mandala and MacKinnon 2022 [↗](#)). After relieving such an obstacle upon depolarization, a rotation of the intracellular ring would allow the helices at the bottom of S6 to dilate, opening the gate (Whicher and MacKinnon 2019 [↗](#); Whicher and MacKinnon 2016 [↗](#); Wang and MacKinnon 2017 [↗](#); Wynia-Smith et al. 2008 [↗](#)). Moreover, cryo-EM data on Kv10.2 (which shows 76% identity with Kv10.1) reveals a pre-open state in which the transmembrane regions of the channel are compatible with ion permeation (the permeation path is dilated, like in open states) but the intracellular gate is still in the same conformation as in closed states (Zhang et al. 2023 [↗](#)). Whether the difference between these two states depends on the rotation of the intracellular ring remains to be addressed.

The intracellular ring is also subject to modulation by factors external to the channel molecule. For example, ligand binding to the PAS domain can modify gating (Wang et al. 2020 [↗](#); Wang et al. 2023 [↗](#)), explaining the complex effects induced by some inhibitors such as imipramine. The association of Kv10.1 with calmodulin (CaM), which efficiently inhibits the channel in the presence of Ca²⁺, adds an additional level of complexity (Whicher and MacKinnon 2016 [↗](#); Schonherr et al. 2000 [↗](#)). Kv10.1 has CaM binding sites at the N- and the C-termini (Gonçalves and Stühmer 2010 [↗](#); Schonherr et al. 1999 [↗](#); Ziechner et al. 2006 [↗](#)). CaM presents two Ca²⁺-binding lobes (N- and C-lobe) joined by a flexible helix. The N-lobe of CaM (Babu et al. 1985 [↗](#)) binds to the N-terminal site in one channel subunit, while the C-lobe binds the C-terminal sites of the opposite subunit (Whicher and MacKinnon 2016 [↗](#)), therefore linking two PAS/CNBHD pairs. It is plausible that his cross-linking of two segments of the ring impairs its rotation and can thus be the basis of the inhibitory action of CaM.

Yet, the mechanism coupling the movement of the voltage sensor and the rotation of the ring remains unsolved. A candidate to fulfill this function is the segment encompassing the first ten N-terminal residues, which, by analogy to Kv11.1, could interact with the distal end of S4 (Stevens-Sostre et al. 2024 [↗](#)). Importantly, mutations in both the intracellular N- (Whicher and MacKinnon 2019 [↗](#)) and C-terminal domains (Zhao et al. 2017 [↗](#)) cause unexpected rectification or paradoxical activation by Ca-CaM instead of the normal inhibition seen in wild type (WT) (Lörinczi et al. 2016 [↗](#)).

In this study, we provide a mechanistic explanation of the hallmarks of Kv10.1 gating and define the role of the intracellular ring using a combination of electrophysiology and mutagenesis, modeling, and biochemical approaches. Channel variants with altered interaction between PASCAP and CNBHD uncovered an open state in the mutants, different from the main open state in the WT. Unlike the canonical open state, the mutant-specific open state is not subjected to a flicker block and, therefore, gives rise to a larger macroscopic current. Because this state is best accessed from deep closed states (strong hyperpolarized membrane potential), it made the transition between the voltage sensor movement and the ring rotation visible, allowing the characterization of gating steps hidden in WT. Furthermore, access to the novel open state is promoted by the binding of CaM, thus explaining the paradoxical effects of CaM in mutant channels. We propose a model tightly constrained by data that can explain the main differential features of Kv10.1 gating.

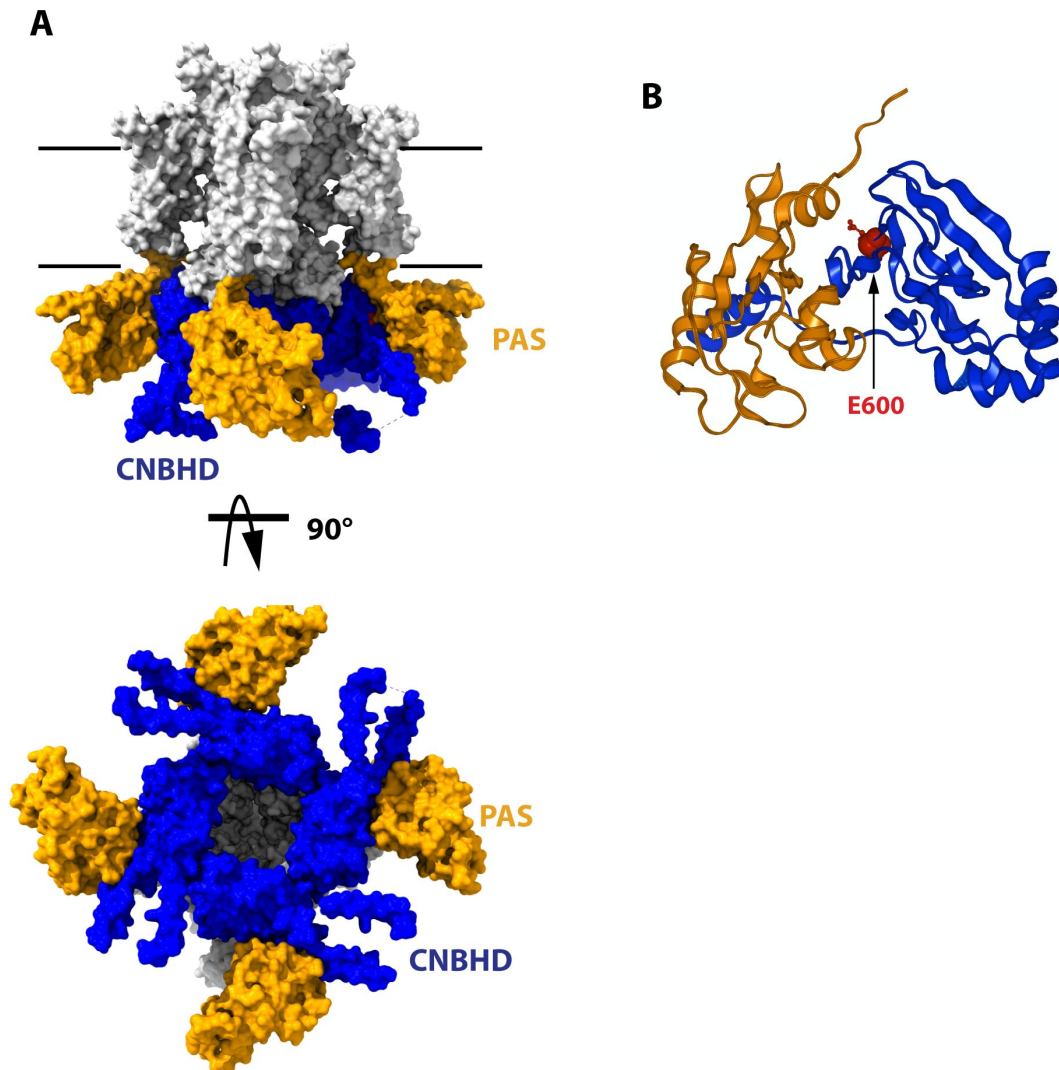


Figure S1.

Structure of the cytoplasmic ring of Kv10.1.

A. Space-filling model from the side and from the cytoplasmic side (bottom). The PAS domains are represented in orange, and the CNBHD is colored in blue. The rest of the protein is drawn in white. Black lines indicate the thickness of the membrane. **B.** Schematic representation of the PAS domain on one subunit and its interaction with the CNBHD of another subunit. The position of E600, mutated in this work, is indicated in red. The structures are based on 8EOW (Mandala and MacKinnon, 2022 [DOI](https://doi.org/10.1016/j.sci.2022.101611)), with the voltage sensor in the uppermost configuration.

Results

Disrupting the interaction between PASCap and CNBHD reveals a biphasic gating behavior

The first N-terminal residues of K_V10.1 (PASCap (Morais-Cabral et al. 1998) are prime candidates for transmitting voltage sensor motion to the intracellular domain based on crystal structures and interactions inferred from mutagenesis experiments. In particular, two residues at the bottom of S4 –H343 (Terlau et al. 1997) and D342 (Tomczak et al. 2017)– functionally interact with the initial N-terminus. The N-terminal PAS domain can then transmit mechanical cues to its C-terminal interaction partner CNBHD (Whicher and MacKinnon 2019; Whicher and MacKinnon 2016; Codding et al. 2020; Codding and Trudeau 2019; Mandala and MacKinnon 2022; Wang et al. 2020). To study K_V10.1 gating under perturbed intramolecular interaction, we deleted the PASCap domain (residues 2-25) and, in a more conservative approach, we examined the impact of a point mutation (E600R, Fig. S1B) reported to disrupt the PASCap-CNBHD interaction (Haitin et al. 2013) and which shows a behavior reminiscent of the PASCap deletion in its response to calmodulin (see below; (Lörinczi et al. 2016)). We then obtained the response of the mutants to discrete depolarizations to different potentials (–100 to +120 mV for 300ms) in *Xenopus* oocytes in the presence of 60 mM K⁺ in the extracellular solution to follow deactivation behavior through tail currents.

Fig. 1A displays representative current traces of WT and the mutant channels. The threshold for activation was shifted towards hyperpolarizing values in both mutants, giving rise to inward currents at potentials that do not lead to the opening of WT channels. This shift is most evident in the biphasic conductance-voltage (GV) plots (Fig. 1B). The kinetics of activation (Fig. 1C) and deactivation (Fig. 1E) of the mutants were also clearly distinguishable from WT. To facilitate the description of the results, we classified the responses into three categories depending on the stimulus amplitude: *weak* (–90 mV to –20 mV), *moderate* (–10 mV to +40 mV), and *strong* (+50 to +120 mV).

While the activation of WT currents does not accelerate dramatically with increasing depolarizations in the voltage range tested, the mutants activated much slower than the WT upon *weak* and *moderate* depolarizations, reaching activation kinetics similar to WT with *strong* ones, as can be observed in normalized traces (Fig. 1C). To obtain a more quantitative estimation of the changes in activation velocity, we used the time required to reach 80% of the maximum current amplitude plotted against the stimulus voltage (Fig. 1D). ΔPASCap and E600R needed a much longer time than WT to activate at *weak* depolarizing potentials but were equally fast at membrane potentials larger than +50 mV.

The deactivation (tail) kinetics changes with increasing depolarizations were more evident and more complex than changes in activation kinetics (Fig. 1E and 1F). For increasing test pulse potentials, the peak amplitude of ΔPASCap tail currents first increased progressively, then decreased at *moderate* values, and rose again after the range of *strong* depolarizations. E600R showed a similar pattern of tail amplitude, while the increase at *strong* potentials was less pronounced although still evident. Remarkably, both the tail amplitude and its decay kinetics underwent profound changes depending on the stimulus. While the kinetics of WT tail currents was the same across different potentials, showing the characteristically fast deactivation of K_V10.1, ΔPASCap deactivated slow- and monotonically at *weak* depolarizations, but a fast component started to become evident after *moderate* stimuli. The fast component dominated the process at *strong* depolarizations. For E600R, the deactivation after *weak* stimuli was also slow and accelerated after a rising phase in the *moderate* and *strong* depolarization range.

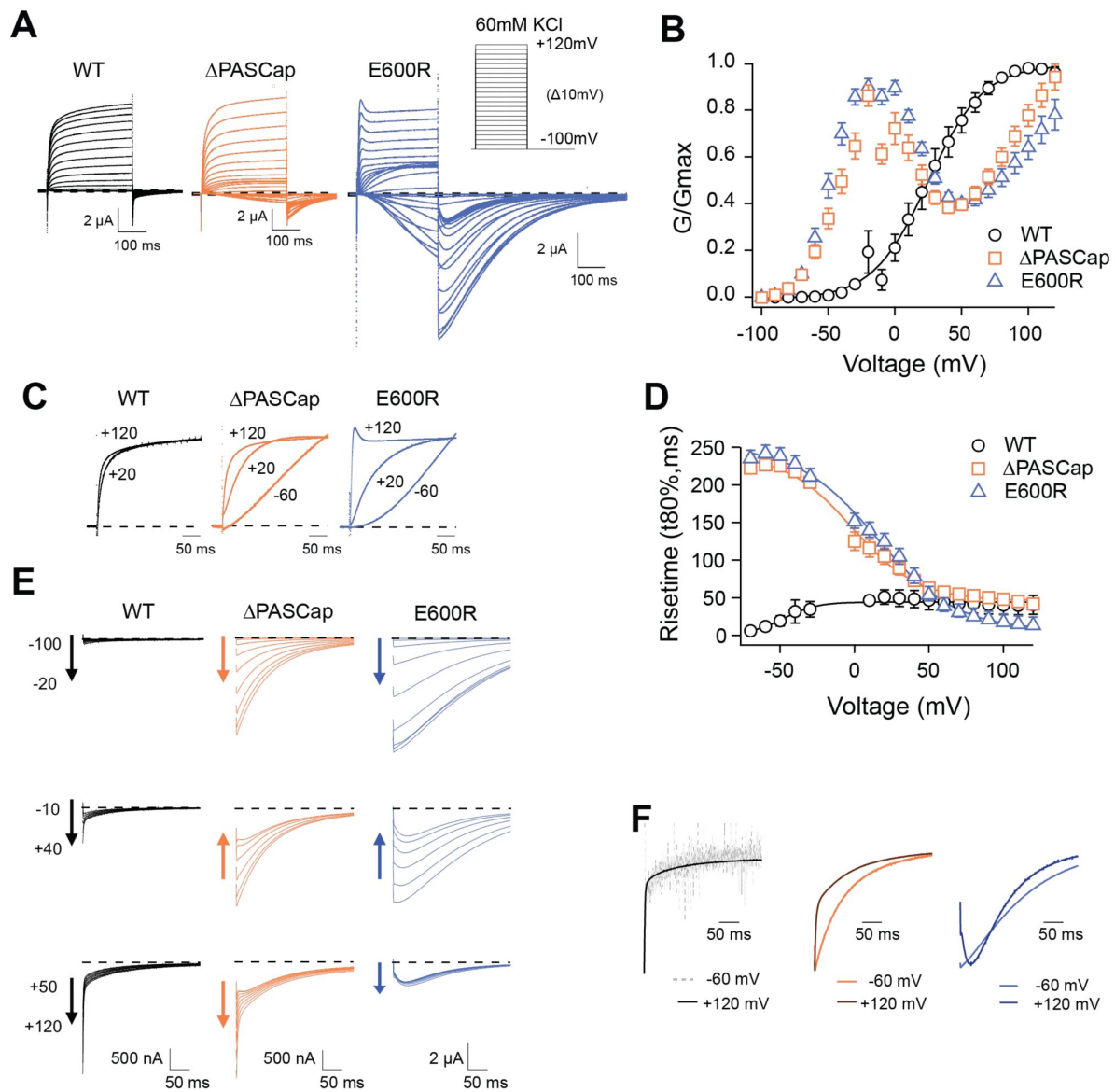


Figure 1.

Characterization of Δ PASCap and E600R mutants.

A. Raw current traces resulting from depolarizations between -100 and +120 mV in WT (black), Δ PASCap (orange), and E600R (blue). The dashed lines indicate zero current. The stimulus protocol is schematically depicted in the inset. The extracellular solution contained 60 mM KCl, 57.5 mM NaCl, 10 mM HEPES, 1.8 mM CaCl₂, pH=7.4. **B.** GV plots corresponding to the three channel variants (colors as in A). (N: WT = 6, Δ PASCap = 7, E600R = 11; mean $\pm$ SEM) **C.** Normalized traces to the indicated voltages to reveal the acceleration of activation with depolarization in the mutants. WT does not give rise to significant outward current at -60 mV. **D.** Rise time of Δ PASCap and E600R as a function of voltage (colors as in A). The activation is much slower than in WT up to +50 mV but reaches the speed of WT with stronger stimuli. **E.** Tail currents at -100 mV after depolarizations to potentials in the weak, medium, or strong range (up to down, see text for details). The arrows indicate the direction of the change in tail peak amplitude with increasing voltage. **F.** Normalized tail currents at -100 mV after depolarizations to the indicated voltages.

Due to the complex behavior of the tail currents, different equations would be needed to fit the tails of the various channels to extrapolate the amplitude to time zero. Hence, to calculate the macroscopic conductance, we simply used the current amplitude at the end of the depolarizing stimulus and divided it by the driving force calculated from the measured reversal potential (**Fig. 1B**). As already observed in the raw traces, the threshold for activation for both mutants was strongly shifted in the hyperpolarizing direction with respect to WT (-80 mV vs. -40 mV). Still, the most evident change was that both Δ PASCap and E600R displayed a biphasic GV in contrast to WT. *Weak* depolarizing pulses increased the macroscopic conductance of both mutants until a maximum at approximately $+10$ mV. With further depolarizations, the conductance initially declined to rise again in response to *strong* depolarizations. This finding matches the changes in amplitude of the tail currents, which, therefore, probably reflect a true change in overall conductance. A similar behavior has been mentioned for related (Whicher and MacKinnon 2019) or unrelated mutations (Zhao et al. 2017) affecting the intracellular domains. However, the reasons for this phenomenon have not been investigated. We thus aimed to understand the molecular mechanisms underlying the biphasic GV.

The biphasic GV is described by two sigmoidal components corresponding to a two-step gating mechanism

One possible explanation for the biphasic behavior could be the coexistence of two separate channel populations with different kinetics, amplitude, and voltage dependence. This seems unlikely because the shape of the GVs was consistent in all our recordings, as evidenced by the small error bars (**Fig. 1B**) despite the variability intrinsic to the oocyte system. Alternatively, each channel could have two open states, and the rectification observed between the two components could represent a transition from one state to the other. Indeed, an equation that reflects the two components and a transition between them (**Equation 4**; see Methods) described the behavior of the GV of both mutants accurately (**Fig. 2A**). Representative current traces are shown in (**Fig. 1A** and **Fig. 2A, C**).

With the available structural information in mind, the two components could represent sequential access to two open states (from here on, O_1 and O_2) through separate gating steps that differentially involve the sensor movement and ring rotation (Tomczak et al. 2017; Whicher and MacKinnon 2019; Mandala and MacKinnon 2022).

To test if the ring underlies one of the two gating steps, we tested the behavior of additional N-terminal deletions of increasing length ($\Delta 2-10$, $2-25$ for Δ PASCap, and $2-135$ for Δ eg), expected to disrupt the ring integrity more and more. A biphasic GV was observed in all these mutants (**Fig. 2A**). The V_{half} value of the first component was very similar across mutants. In contrast, the second component showed different thresholds. We then performed a global fit using **equation 4**, where we linked the parameters of the first components (V_{h1} , K_1) across mutants and allowed the parameters for the second component (V_{h2} , K_2 , A_2) and the transition (V_{h3} , K_3) to vary (**Table S1**, **Fig. 2B**). The global fit accurately described the behavior of the GV in all mutants (**Fig. 2A**).

The global fit results indicate that the first component is conserved across mutants. In contrast, the second component occurs at progressively more depolarized potentials for increasingly larger N-terminal deletions and when the structure of the ring is altered through disruption of the interaction between N- and C-termini (E600R). Thus, the underlying gating events can be separated into two steps: The first gating step involves only the voltage sensor without engaging the ring and leads to a pre-open state, which is non-conducting in the WT but conducting in our mutants. The second gating event operates at higher depolarizations, involves a change in the ring, and leads to an open state both in WT and in the mutants.

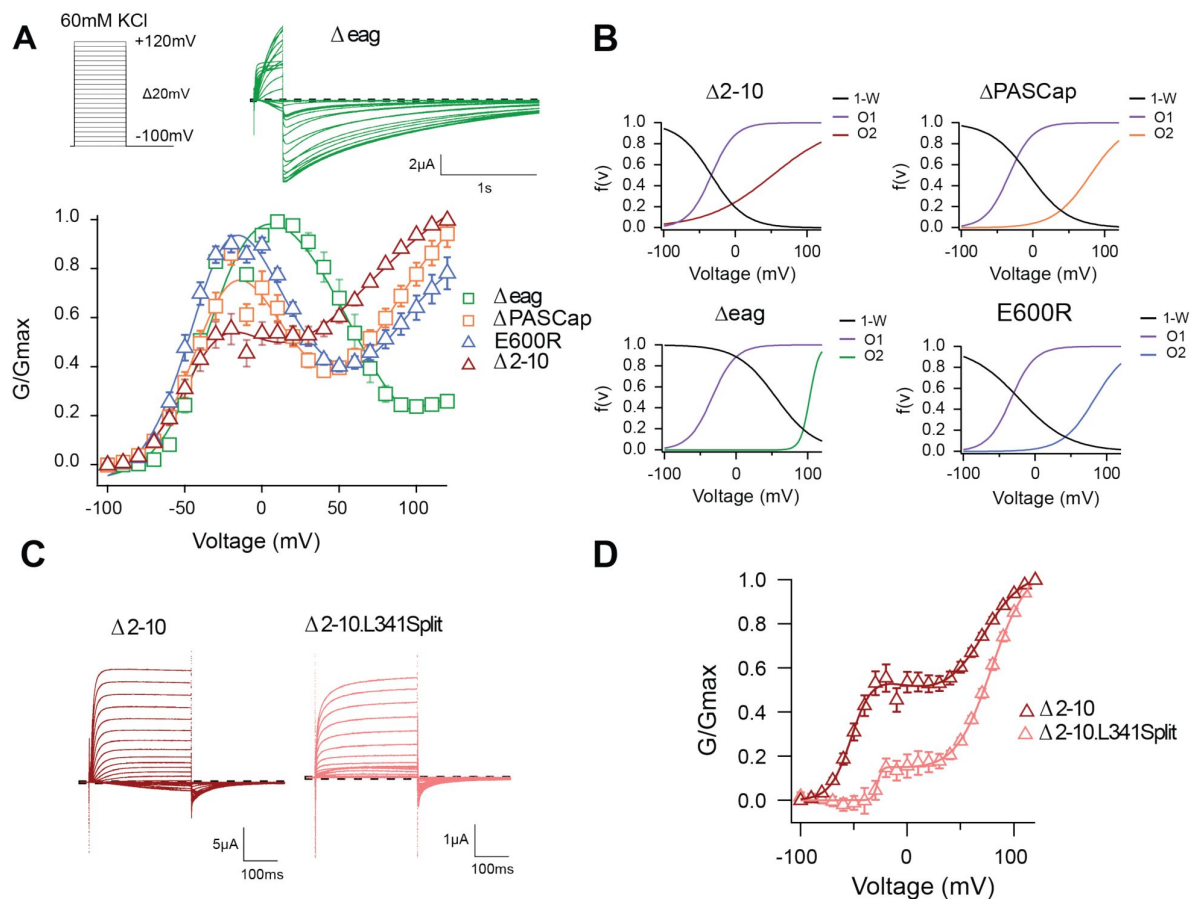


Figure 2.

The biphasic GV corresponds to two sequential events.

A. The GV of all tested mutants show biphasic behavior. (N: $\Delta 2-10$ = 6, $\Delta PASCap$ = 7, Δeag = 7, E600R = 11; mean $\pm$ SEM). All are well described by [Eq. 4](#) in a global fit with fixed parameters for the first component. **B.** Distribution of the three components used for the fits as a function of voltage. **C.** A discontinuous form of $\Delta 2-10$ shows attenuated biphasic behavior. **D.** GV plots of $\Delta 2-10$ and $\Delta 2-10.L341Split$ (N: $\Delta 2-10$ = 6, $\Delta 2-10.L341Split$ = 5; $\pm$ SEM). Split fitted using [Eq. 4](#). The dashed lines in **A** and **C** indicate zero current. The stimulus protocol is schematically depicted in the inset in **A**. The extracellular solution contained 60 mM KCl.

	$\Delta 2-10$	ΔPASCap	Δe_{ag}	E600R
A_0	-0.052	-0.054	-0.067	-0.091
A_1	1.78	1.78	1.32	2.99
Vh_1	-33.57	-33.57	-33.57	-33.57
K_1	16.38	16.38	16.38	16.38
A_2	1.31	1.18	0.25	0.99
Vh_2	51.63	79.63	102.70	83.04
K_2	46.05	25.38	6.41	22.51
Vh_3	-33.47	-5.16	55.52	-21.93
K_3	23.47	27.12	27.01	34.9

Table S1.

Parameters of a global fit that linked the first component of the biphasic response.

To test this hypothesis, we compared the behavior of $\Delta 2-10$ and $\Delta 2-10.L341Split$, a channel lacking a covalent connection between the sensor and the pore domain (Tomczak et al. 2017), hence decoupling residues downstream S4 (in this case starting from 342) from the movement of the sensor. This decoupling would minimize the action of the displacement of the voltage sensor on the pore domain but would not necessarily affect the subsequent ring rotation. As predicted, compared to $\Delta 2-10$, $\Delta 2-10.L341Split$ showed a significant reduction in the first component of the biphasic GV (Fig. 2C, D). This indicates that the coupling of the voltage sensor motion to the entry into O_2 does not require a continuous S4-S5 linker. If, in non-split mutants, the upward transition of S4 allows entry to O_1 , it is reasonable to assume that the movement is not transmitted the same way in the split and the transition into O_1 is less probable. The observation that, in the split, entry into O_1 requires higher depolarization and appears to be less likely, suggests that downstream of S4 (beyond position 342), there is a mechanism to convey S4 motion to the gate of the mutants.

Activation of WT $K_v10.1$ channels (best studied for the *Drosophila* form *eag*) drastically slows down in the presence of extracellular divalent cations that bind to the voltage sensor (Silverman et al. 2000; Tang et al. 2000), e.g., Mg^{2+} , Ni^{2+} , Co^{2+} , and Mn^{2+} , but not Ca^{2+} . Strikingly, the degree of deceleration correlates with the ions' hydration enthalpy, suggesting that the ion might unbind during activation (Terlau et al. 1996). We chose to study the impact of Mg^{2+} on the kinetics and voltage dependence of activation (Fig. 3) in the mutant Δeag , which shows the most significant separation between the two conductance components.

Mg^{2+} slowed the activation of the channel at all depolarizations, affecting entry to both open states. This result suggests that Mg^{2+} also slows the voltage sensor motion that controls access into the second open state. In addition, the voltage dependence of activation was shifted by approximately 25 mV to more depolarized voltages. This is in line with deep deactivated states bound to Mg^{2+} and requiring stronger depolarization to exit them. Interestingly, the transition from the first to the second component of the GV plot seems unaffected by the extracellular divalent binding, which could be expected if the transition occurs in the cytoplasm.

Steady-state voltage dependence and activation kinetics are consistent with two open states (O_1 and O_2) with different conducting behavior

Thus far, our results are compatible with two different open states in the mutants. The first open state, the mutant-specific O_1 , would dominate at *weak* depolarizations and deactivate slowly, with tail current decay time constants of tens of milliseconds. In contrast, mutant and WT channels would reach the second open state (O_2) at strong depolarizations and deactivate more rapidly, with a few milliseconds or less time constants. A critical test of this hypothesis is the application of more complex voltage stimuli that drive the system to a non-equilibrium state of high O_1 occupancy. This might be achieved by deactivation periods of around 10 ms, just long enough to remove most channels from O_2 but sufficiently brief to maintain the occupancy of O_1 . For simplicity, we used depolarization periods of the same duration and tested whether a 300 ms series of activating and deactivating 10 ms pulses could accumulate channels in a high conducting state O_1 .

For WT (Fig. 4A), alternating between -80 and +80 mV resulted in a smaller amplitude than a constant stimulus to +80 mV, just as expected for a system with a single open state and a monotonic voltage dependence of activation. Moreover, WT's rapid deactivation resulted in near-complete deactivation during every cycle.

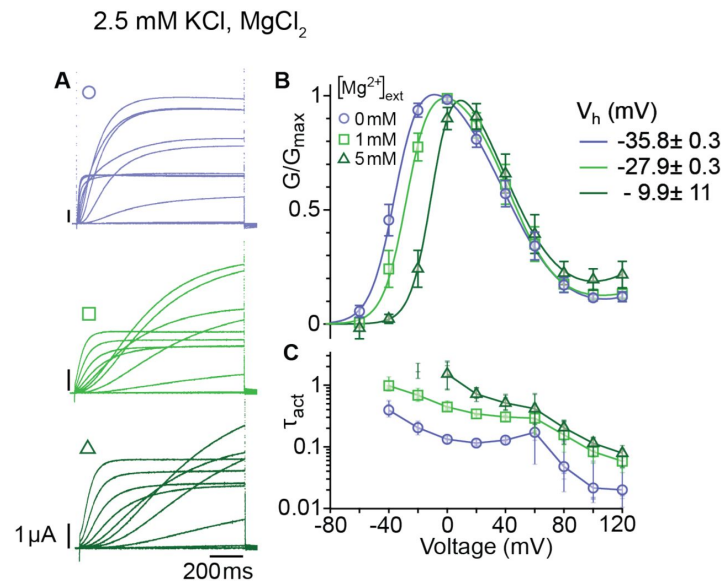


Figure 3.

Mg²⁺ induces a shift of the first component in the depolarizing direction.

A. Raw current traces from oocytes expressing Δeag channels in response to depolarizations from a holding potential of -100 mV to voltages between -100 and +120 mV in the presence of 0, 1 or 5 mM MgCl₂ in the external solution (legend in B). Scale bars are 1 μ A and 200 ms. **B.** Average GV plots (with SEM) obtained under the same conditions from n=6, 6, and 5 recordings (0, 1, and 5 mM MgCl₂). The V_{half} of the first component shifts from -35.8 mV in the absence of extracellular Mg²⁺ to -27.9 and -9.9 mV in 1 mM and 5 mM MgCl₂. **C.** Time constants of the early activation show the decelerating effect of Mg²⁺. Symbols depict averages, vertical lines the range and horizontal lines individual experiments. For voltages where conductances or time constants could not be reasonably estimated no data are displayed. The recordings were performed in an external solution containing 115 mM NaCl, 1.8 mM CaCl₂, 2.5 mM KCl, 10 mM Hepes pH 7.2, and the indicated concentrations of MgCl₂,

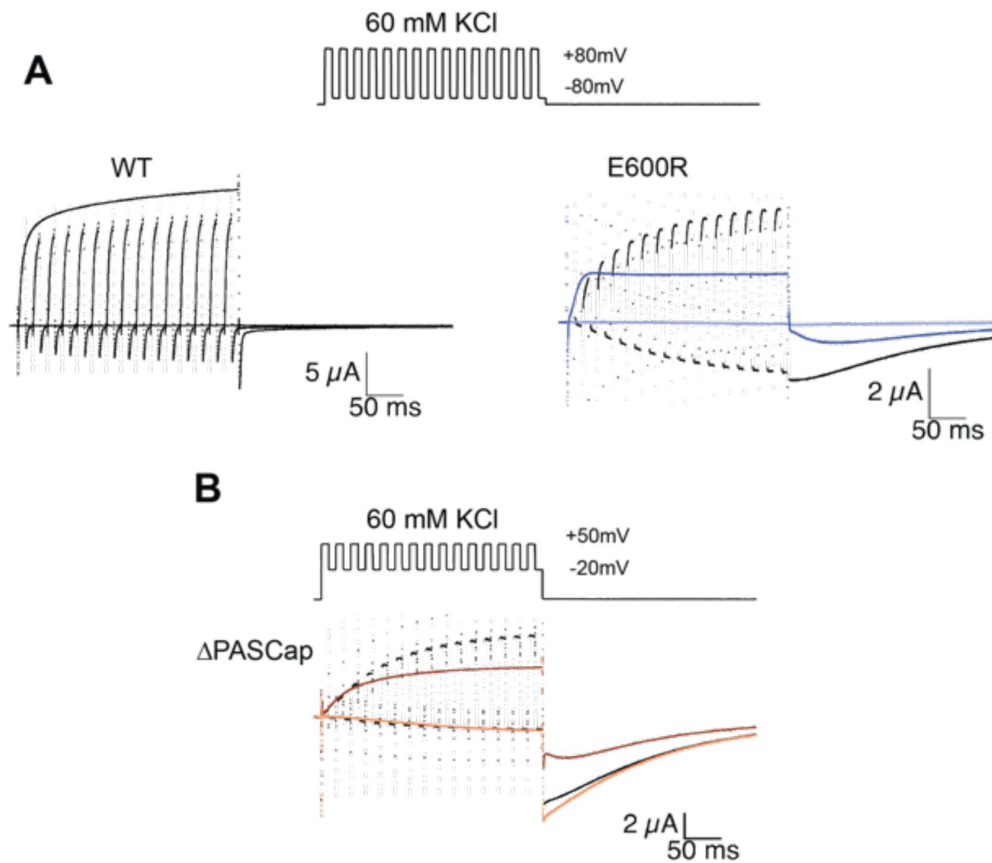


Figure 4.

Alternating stimuli reveal larger macroscopic conductance for O1.

A. Alternating potential between -80 and +80 mV in the WT results in current amplitudes that are smaller than those during a sustained stimulus (Upper left traces). In contrast, E600R gave rise to larger currents when the stimulus was intermittent and too short to allow occupancy of O2 (upper right). Current traces in response to a sustained stimulus are shown in blue (dark trace +80 mV ; light trace is -80 mV). **B.** The effect was qualitatively similar for Δ PASCap, which consistently gave rise to larger current upon oscillating stimuli between -20 and +50 mV than during a constant pulse to +50 mV. Current traces in response to a sustained stimulus are shown in orange (dark trace +50 mV ; light trace is -20 mV). The stimulus protocol for A and B is schematically depicted in the inset. The extracellular solution contained 60 mM KCl.

In E600R, in contrast, the current amplitude during activating pulses increased steadily from cycle to cycle. Ultimately, the current amplitude exceeds that obtained with a constant +80 mV pulse (Fig. 4A). Because the deactivation of this mutant is much slower and occurs at more negative potentials than that of WT (see Fig. 1 and 2B), there was no noticeable deactivation during the -80 mV episodes. In addition, the tail current after the alternating stimuli had a larger amplitude than the one after a continuous pulse and decayed mostly as one component. All those observations are consistent with an accumulation of channels in O_1 . A similar behavior was detected with Δ PASCap (Fig. 4B), albeit within a different voltage range, as could be predicted from the different voltage dependence of the transition between states (Fig. 2B). Again, alternating pulses resulted in larger current amplitudes and less complex tail current as compared to a single long pulse (Fig. 4B).

Single channel recordings reveal two distinct voltage-dependent behaviors in Δ PASCap

A larger current amplitude in O_1 can be due to a larger single-channel conductance or a different open probability. We performed single-channel recordings in membrane patches excised from oocytes expressing the Δ PASCap mutant to distinguish between these two possibilities. To limit the possible contamination by other channels, we measured the activity in outside-out patches with K^+ as the only cationic charge carrier on the cytoplasmic side (see Methods), rendering a theoretical Nernst equilibrium potential for K^+ of -93.77 mV and -4.69 mV for Cl^- . We only considered channels producing outward currents at -60 mV, ruling out Cl^- contamination. In such patches, we detected K^+ current during single-channel openings that typically lasted several milliseconds upon weak depolarizations but could rarely be resolved at strong depolarizations (Fig. 5A). Such behavior was never detected in uninjected oocytes. The activity in the patches was silenced by the addition of 100 μ M astemizole, providing robust evidence that the activity is the result of the expression of Δ PASCap.

To characterize the activity in a quantitative way, we performed series of sweeps where the patch was held at -100 mV for 10 s, depolarized to +10 mV for 2.5 s, held again at -100 mV for 5 s, and finally depolarized to +40 mV for 2.5 s. The procedure was repeated tens of times on each patch, and representative traces are shown in Fig 5B. At +10 mV, we detected openings lasting up to hundreds of milliseconds, while the dwell times were much shorter at +40 mV. In some cases, openings of the two classes were observed in the same sweep, with the long openings preferentially preceding the shorter ones. Any transition from short openings to long openings (or back) always involved a longer intermediate close time, indicating that the channel travels between the two states through one or more closed states without a direct link between the two open states.

Deep-closed states favor access to O_1

A hallmark of $K_v10.1$ gating is the change in activation kinetics in dependence on the pre-pulse potential reminiscent of (and commonly named) the Cole-Moore Shift (Cole and Moore 1960; Hoshi and Armstrong 2015). Hyperpolarized potentials drive the channel into deep closed states, which delays and decelerates activation (Ludwig et al. 1994). This is well described by a model with four identical, independent transitions of the voltage sensor (Schönherr et al. 1999) and is compromised by N-terminal deletions (Whicher and MacKinnon 2019). To test the behavior of our mutants concerning this property, we applied a series of 5s-long conditioning pulses with voltages ranging from -160 mV to -20 mV, followed by a test pulse to +40 mV in the absence of external Cl^- . The activation kinetics, quantified by the time to 80% of the maximum current, showed a strong pre-pulse dependence in WT, Δ PASCap, and E600R, with much larger rise times in both mutants (Fig. 6B).

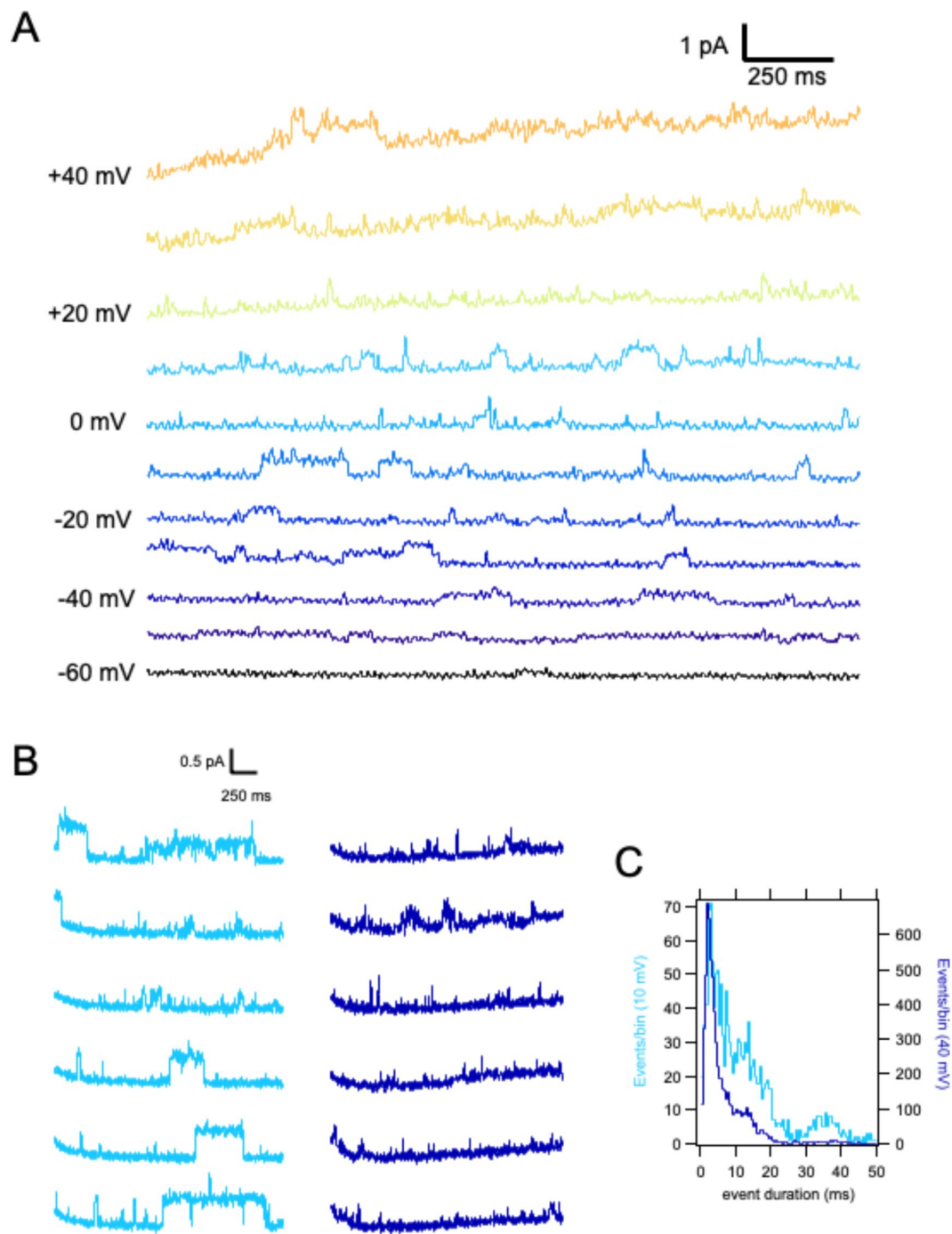


Figure 5.

Single Δ PASCap channels reveal longer open times at moderate depolarizations.

A. representative current traces at the indicated voltages obtained from a holding potential of -100 mV. **B.** Comparison between traces obtained at +10 mV (light blue) and +40 mV (dark blue). The time spent in the open state is much longer at +10 mV. **C.** Open time distribution for events shorter than 50 ms obtained from currents in an experiment like the one depicted in B, (100 stimuli at each potential).

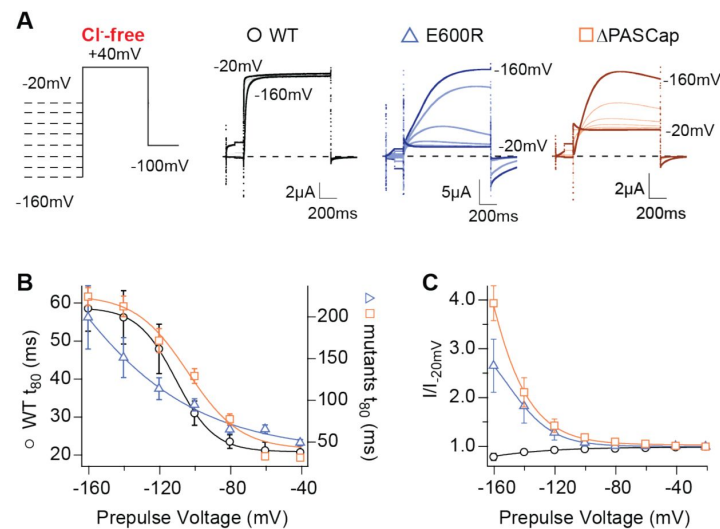


Figure 6.

Hyperpolarization promotes access to a large conductance, slowly activating open state.

A. Raw current traces in response to the stimuli depicted in the scheme. **B.** The rise time to 80% of the maximal current during the depolarizing stimulus is plotted vs. prepulse voltage (N: WT = 7, ΔPASCap = 9, E600R = 8; mean±SEM). Although the activation is much slower for both mutants (note the different y axis for the mutants), they retain a strong dependence on the prepulse potential. **C.** Normalized end-pulse current (I/I_{-20}) is plotted vs. prepulse voltage (N: WT = 10, ΔPASCap = 8, E600R = 8; mean±SEM). The amplitude of the current at +40 mV increased markedly when the holding potential was below -100 mV in the mutants, while the amplitude in WT changed only marginally. The dashed lines indicate zero current. The stimulus protocol is schematically depicted in the inset. A 5s pre-pulse was applied, of which only the initial and last 100ms were recorded. The chloride-free extracellular solution contained 115 mM Na-methanesulfonate, 2.5 mM KOH, 10 mM Hepes, 1.8 mM Ca(OH)₂, pH=7.2.

In the mutants, not only activation kinetics but also current amplitude was substantially affected by hyperpolarizing pre-pulses. With respect to the -100 mV pre-pulse potential, the current starting from a -160 mV pre-pulse increased in Δ PASCap by a factor of 3.93 ± 0.36 and in E600R by a factor of 2.65 ± 0.54 . In contrast, WT current amplitude was not significantly affected (**Fig. 6A** [↗](#), 6C). Such increases in amplitude are often related to augmented channel availability due to voltage-dependent recovery from inactivation. Still, although under some conditions, a decrease in outward current can be observed during the stimulus (see, e.g., traces of E600R in **Fig. 1** [↗](#) and traces in **Fig. 6** [↗](#)), we never detected in any mutants cumulative inactivation after repeated depolarization, as depicted, e.g., in **Fig. 4** [↗](#). Transition from a conducting to a non-conducting state while the depolarization persists is the definition of inactivation. We nevertheless prefer to refer to the non-conducting states as “closed” rather than “inactivated”, because O_1 is non-conducting (or not accessed) in the WT, and therefore, the subsequent states cannot be considered inactivated. The state resulting from the flickering block from O_2 is arguably inactivated, but such events are commonly not defined as inactivated because of their unstable nature.

In the absence of canonical inactivation, the prepulse-dependent current increase at +40 mV could be related to changes in the relative occupancy of the open states. We hypothesized that the open state O_1 , which produces larger macroscopic currents, might be more accessible after a hyperpolarization. The current decay after the peak (**Fig. 6A** [↗](#)), especially in Δ PASCap, also indicates a transient phenomenon. To map the voltage dependence of this effect more comprehensively, we next compared the effect of hyperpolarized pre-pulse on currents elicited at different test potentials.

We tested the effect of pre-pulse potentials (-160 and -100 mV) on IV protocols in the absence of Cl^- . Compared to a -100 mV pre-pulse, -160 mV clearly potentiated the first component of Δ PASCap and E600R biphasic IV (**Fig. 7A** [↗](#) and **7B** [↗](#)).

If the hyperpolarizing potentials facilitate the access to O_1 by driving the channel into deep closed states, then impairing the access to these states will reduce the component corresponding to O_1 in the GV. The mutation L322H, located in the S4 segment, limits access to deep closed states in WT channels (Schönherr et al. 1999 [↗](#)). We introduced this mutation in the context of Δ PASCap and E600R. Representative current traces obtained from Δ PASCap^{L322H} and E600R^{L322H} are shown in **Fig. 7C** and **D** [↗](#). Both mutants showed drastic attenuation in the first component of the biphasic GV compared to the parental channels. The tail currents of Δ PASCap^{L322H} and E600R^{L322H} did not show rectification. They presented homogenous kinetics at all potentials, indicating that reducing the access to deep closed states also reduces the occupancy of O_1 .

Ca²⁺ Calmodulin stabilizes O_1

The available cryo-EM structures show K_v10.1 in a complex with Ca²⁺-CaM and therefore in a non-conducting state, as it is well established that binding a single Ca²⁺-CaM inhibits the current through WT channels (Schönherr et al. 2000 [↗](#); Ziechner et al. 2006 [↗](#)). In stark contrast, increasing intracellular Ca²⁺ has been reported to potentiate Δ PASCap and E600R current amplitudes (Lörinczi et al. 2016 [↗](#)). In the light of the results described above, this seemingly paradoxical behavior of mutants could be explained if Ca²⁺-CaM binding restricts the access to the WT-like open state O_2 while facilitating access to O_1 , which gives rise to larger currents and is not visible in WT channels. Therefore, we predicted that Ca²⁺-CaM binding would potentiate the first component of the biphasic IV in the mutants.

To test this, we induced a rise in cytosolic Ca²⁺ using the ionophore ionomycin and inducing release from the stores with thapsigargin (both 5 μ M) (Lörinczi et al. 2016 [↗](#)). Because the increase in intracellular Ca²⁺ is transient and reverts after approximately 150s (see Supp. 1 to Fig. 8), and our protocols with discrete voltage pulses require a long time to complete, we used a 5s voltage ramp from -120 to +100 mV repeated every 30s for 300s. The currents were recorded in Cl^- -free

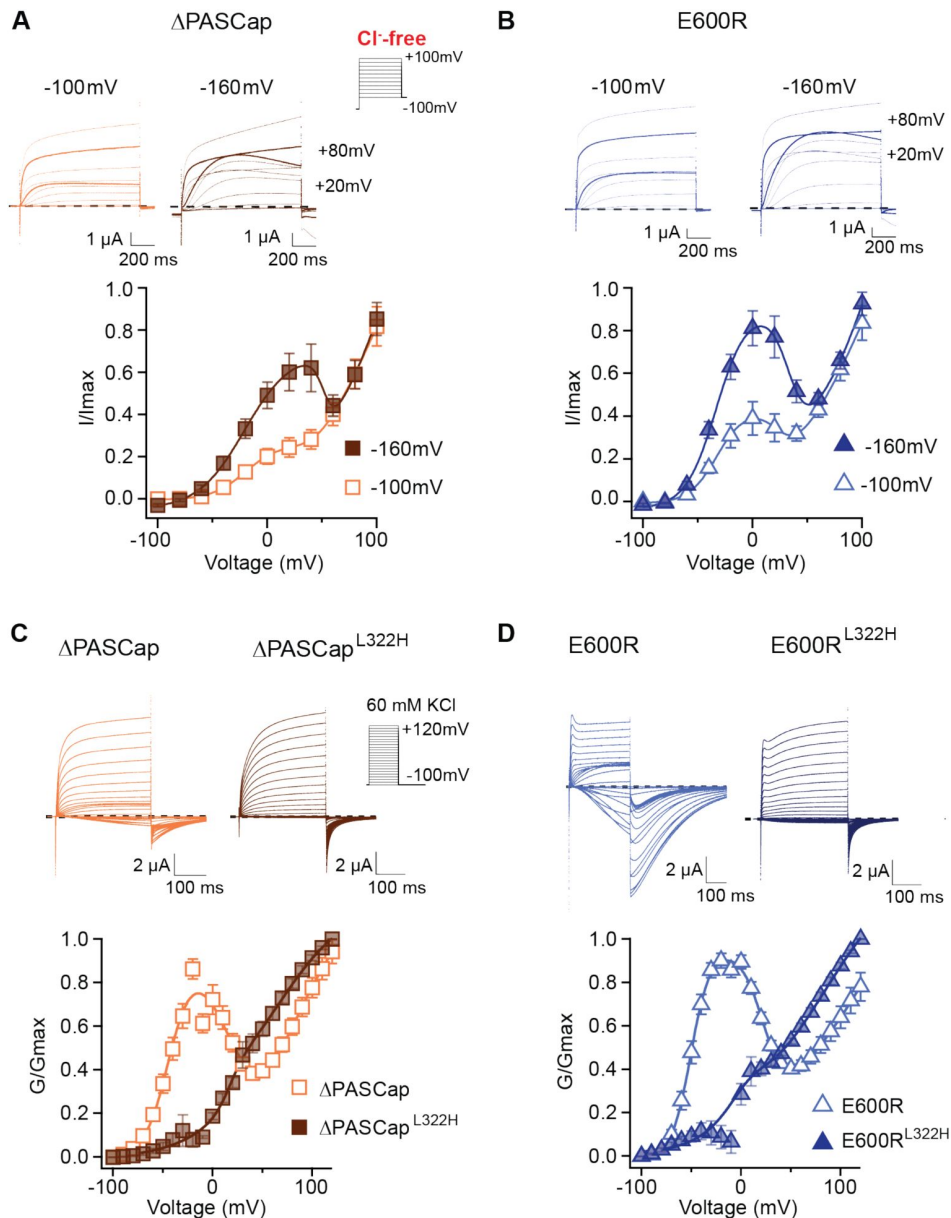


Figure 7.

Deep closed states facilitate access to O1.

A, B. Conditioning pulses to -160 mV potentiated the first component and hence the biphasic behavior of the I/V relationships for Δ PASCap (A) and E600R (B). (N: Δ PASCap = 7, E600R = 6; mean $\pm$ SEM) The stimulus protocol is schematically depicted in the inset. A 5s prepulse to -160 or -100 mV was applied, which is not shown in stimulus cartoons or current traces. The extracellular solution did not contain chloride. **C, D.** A mutation known to impair access to deep closed states (L322H) largely removes the initial phase of the GV curves for Δ PASCap (C) and E600R (D). (N: Δ PASCap = 7, Δ PASCap^{L322H} = 6, E600R = 11, E600R^{L322H} = 7; mean $\pm$ SEM). Recordings were performed in a solution containing 60 mM KCl. The stimulus protocols are depicted schematically in the insets. The dashed lines indicate zero current.

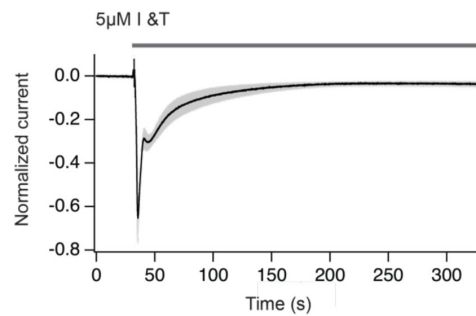
extracellular solution to avoid confounding effects of the endogenous Ca^{2+} -dependent Cl^- channels. The results of representative experiments on ΔPASCap and E600R are shown in the upper left panels of **Fig. 8A and B**, respectively. 60 s after ionomycin/thapsigargin application, a marked potentiation of the current amplitude was observed, and the response to the ramp became linear. The current amplitude declined over the following 300s, and the biphasic IV characteristics partly recovered. The speed and extent of recovery were higher for ΔPASCap than E600R; after 150s, the first component recovered by $39.84 \pm 6.35\%$ for ΔPASCap and $11.84 \pm 3.07\%$ for E600R. This recovery time course matches that of the changes in cytoplasmic Ca^{2+} and agrees with the results obtained by Lörinzi et al. for constant pulses (Lörinzi et al. 2016). The magnitude of potentiation and change in IV shape was homogeneous enough among oocytes to allow averaging of the normalized current in all experiments (**Fig. 8A and B**, lower panel). The changes in slope can be easily observed in the corresponding first derivative of the normalized IV as a function of voltage shown in Supp. 2 to Fig. 8.

The changes in amplitude and kinetics in response to rising intracellular Ca^{2+} support our hypothesis that Ca^{2+} -CaM stabilizes O_1 , possibly by driving the channels to deep closed states (**Figs. 7** and **8**), which would result in smaller currents in WT and larger currents in mutants with access to O_1 . We, therefore, predicted that forcing the channels to deep closed states using Ca^{2+} -CaM could restore access to O_1 in the channels with restricted access to deep closed states $\Delta\text{PASCap}^{\text{L322H}}$ and $\text{E600R}^{\text{L322H}}$. This was tested with the approach described above, increasing intracellular Ca^{2+} while recording a repeated ramp protocol in $\Delta\text{PASCap}^{\text{L322H}}$ and $\text{E600R}^{\text{L322H}}$. As seen in the representative traces in the upper right panels in **Fig. 8A and B**, we observed a notable increase in current 60 s after Ca^{2+} rise, limited to *moderate* potentials, which results in a biphasic behavior resembling that of the ramps in the parental mutants in the absence of Ca^{2+} . For both mutants, the IV returned to a linear shape after 300s. Consistent with the observations for ΔPASCap and E600R, $\Delta\text{PASCap}^{\text{L322H}}$ exhibited faster recovery than $\text{E600R}^{\text{L322H}}$. Traces normalized to the maximum current and averaged are shown in **Fig. 8A and B** (lower panel). The respective first derivatives are shown in Supp. 3 to Fig. 8.

To further explore the role of CaM, we introduced mutations to preclude CaM binding at the N-terminal (F151N L154N) and C-terminal (F714S F717S) binding sites. We tested the behavior of these mutants (termed BDN and BDC2) in the context of ΔPASCap and E600R constructs (**Fig 8C and D**) under conditions of basal intracellular Ca^{2+} concentration. $\Delta\text{PASCap}^{\text{BDN}}$ showed a GV relationship like ΔPASCap . In contrast, $\Delta\text{PASCap}^{\text{BDC2}}$ lost its biphasic behavior. In the case of E600R, mutation of the C-terminal binding site ($\text{E600R}^{\text{BDC2}}$) showed attenuation of the first component of the GV, although not complete. The attenuation of the first component when binding of CaM to its C-terminal binding site is disrupted supports our hypothesis that Ca^{2+} -CaM stabilizes O_1 . The significance of CaM for the mutants seems different. CaM might be crucial for stabilizing O_1 in ΔPASCap while less critical in E600R.

These results opened the possibility that the biphasic behavior is due to two coexisting populations of channels, depending on CaM binding at rest. We considered this possibility, but we find it is unlikely because i) the behavior of the mutants is very homogeneous among oocytes, in which resting Ca^{2+} is very variable; it can be as high as 400 nM (Busa and Nuccitelli 1985), although ten times lower concentrations have also been reported (Parker et al. 1996) and ii) two independent populations of channels would not explain the rectification or the voltage-dependent transitions during short repeated alternating stimuli.

Still, it was possible that CaM is permanently bound to the channel and participates in the gating machinery while only inhibiting the current when bound to Ca^{2+} . Although the available literature would not be compatible with this hypothesis (Schönherr et al. 2000; Ziechner et al. 2006), we estimated the fraction of channels (WT or mutant) bound to CaM as a function of free Ca^{2+} concentration. We co-expressed Myc-tagged CaM and $\text{K}_v10.1$, extracted the proteins in different Ca^{2+} concentrations (see Methods), and pulled down the complex through the Myc-tag of CaM.



Supplement 1 to Figure 8.

Chloride currents obtained upon treatment with ionomycin plus thapsigargin. The increase in cytosolic Ca^{2+} translates in a Cl^- current that can be used to estimate the amplitude and duration of the Ca^{2+} increase. Notice that Ca^{2+} returns to basal levels in approximately 150 s. The gray area indicates SEM. N=9

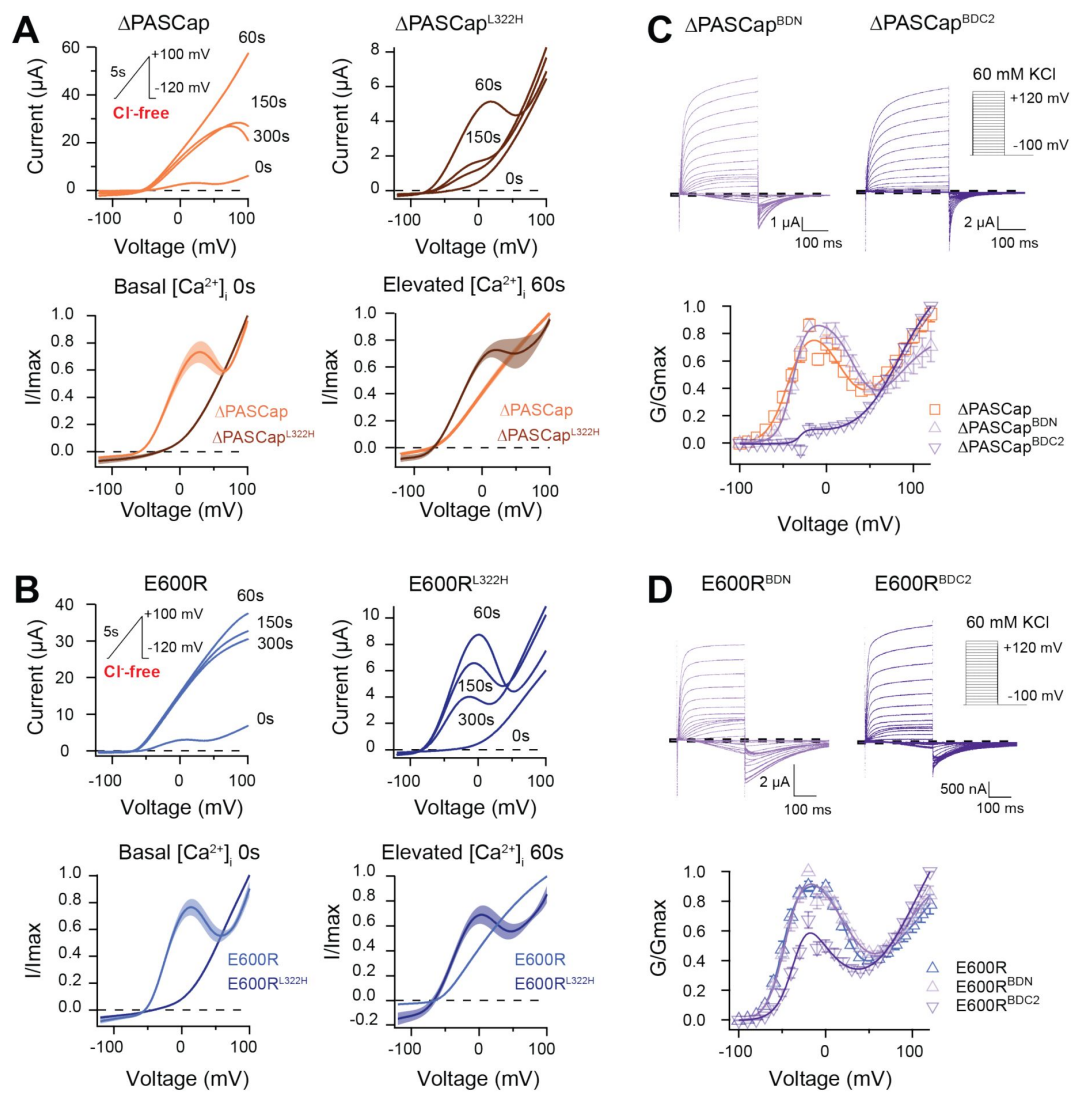


Figure 8.

CaM stabilizes O1.

A. A transient rise in intracellular Ca^{2+} using 5 μ M ionomycin and thapsigargin increases Δ PASCap current amplitude (in the absence of external chloride) and the IV relationship becomes linear (upper left traces). The same treatment in a channel carrying a mutation (L322H), reducing access to deep closed states, results in the appearance of a biphasic IV upon Ca^{2+} rise (upper right traces). The lower traces represent the average normalized response comparing 0 and 60s. (Δ PASCap, light trace, N = 8, Δ PASCap^{L322H}, dark trace, N = 5; the shadowed area indicates SEM). **B.** E600R behavior is comparable to Δ PASCap. Average normalized traces (E600R, light trace, N = 10; E600R^{L322H}, dark trace, N = 11; the shadowed area indicates SEM) are represented in the lower panel. The ramp protocol for A and B is schematically represented in the inset. The extracellular solution contained 2.5 mM K^+ and no chloride. The dashed lines indicate zero current level. **C.** Mutation of the C-terminal CaM binding domain (BDC2) in Δ PASCap strongly reduces the first component of the biphasic GV, while deletion of the N-terminal binding site (BDN) did not have any effect (N: Δ PASCap = 7, Δ PASCap^{BDN} = 7, Δ PASCap^{BDC2} = 7; mean $\pm$ SEM). **D.** The reduction of the first component in E600R when the C-terminal CaM binding site is mutated is also present but less intense. (N: E600R = 11, E600R^{BDN} = 9, E600R^{BDC2} = 11; mean $\pm$ SEM). The experimental protocols for C and D are represented in the insets. Currents were recorded in the presence of 60 mM KCl. The dashed lines indicate zero current level.

Under these conditions, we could consistently detect a fraction of channels bound to CaM already in 50 and 100 nM Ca^{2+} , which increased dramatically in 0.5 and 1 μM , compatible with previous reports, suggesting that, at basal Ca^{2+} , only a small and variable fraction of channels are bound to CaM both in the WT and in the mutants (Supp. 4 to Fig. 8). The fraction of channels bound to CaM at basal Ca^{2+} (in the range of 4% in 100 nM), even if only one CaM is needed per channel tetramer, is insufficient to explain the prominent and proportionally constant first component observed in the mutants.

In summary, our results strongly suggest that Ca^{2+} -CaM stabilizes O_1 , possibly by driving the channel to deep closed states.

A two-layer Markov model recapitulates the kinetic features of ΔPASCap

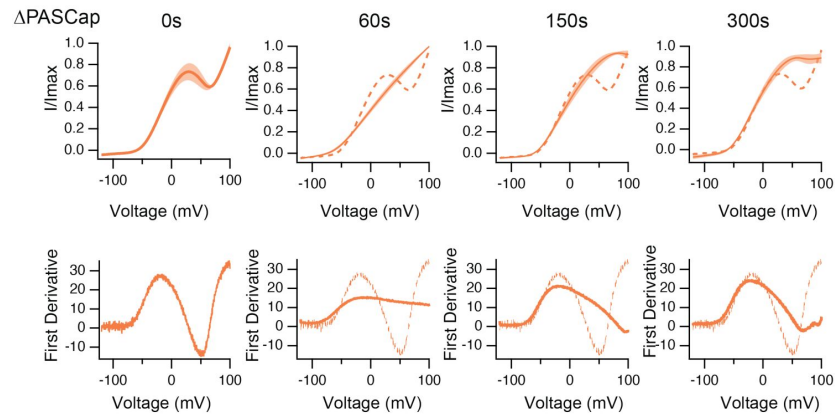
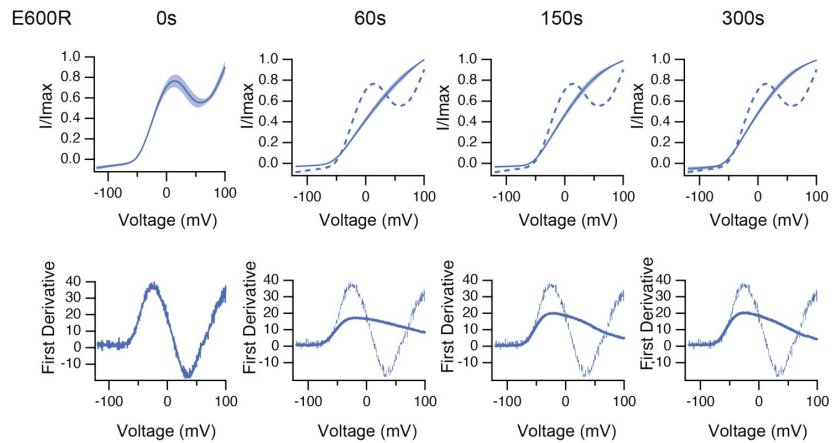
So far, our experimental results suggest that an additional open state exists in $\text{Kv}10.1$ mutants with a compromised intramolecular coupling. This hypothesis can explain the biphasic GV curves, the tail currents' complex shape (Fig. 1 and 2), the current increase following repeated brief hyperpolarizations (Fig. 4), and even the paradoxical current increase under rising intracellular Ca^{2+} concentrations (Fig. 8 and its supporting figures). However, in each case, there might be alternative explanations, such as two separate channel populations with different properties or voltage-dependent de-inactivation. The ultimate test for the hypothesis is the construction of a single model that consistently and quantitatively captures all these unusual characteristics.

It should be noted that the model structure presented here (Suppl. 1 to Fig. 9) is not the only one we found to be able to reproduce the data, but it is amongst the simplest that could (see Methods).

Given the large number of model parameters (42+absolute conductance), it might be surprising that all parameters can be constrained. However, the wide range of voltage protocols and the concurrent matching of depolarization and repolarization responses tightly constrains several rate ratios at different voltages and, thereby, ultimately all parameters. This said, the model never fitted all experiments at once very well. Consequently, a global fit proved extremely hard to achieve in an automated way, and we decided to explore the parameter space manually, based on the time constants and rate ratios we could discern from the experiments.

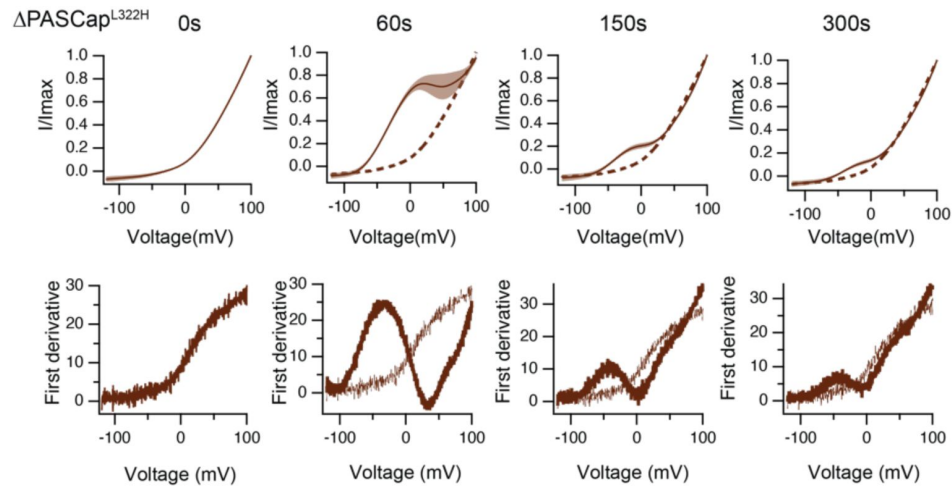
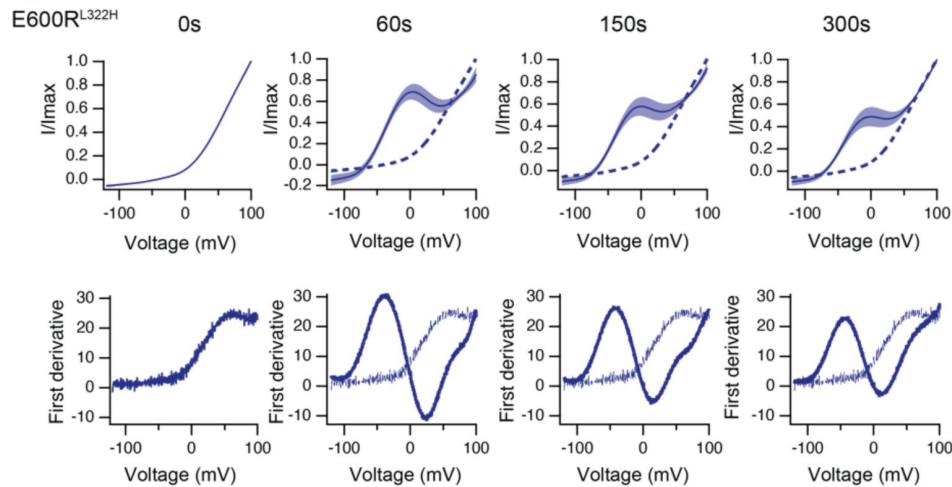
The model parameters are listed in Table S2. These parameters and the structure in Supplement 1 to Fig. 9 jointly define the model that produces all results in Fig. 9 and its supplementary figures for ΔPASCap . Modifications are only necessary to capture the effect of Ca^{2+} -CaM-binding and the properties of other channel variants. Suppose the binding influences the interaction between the intracellular modules that stabilize different configurations of the gating ring, e.g., BDC2, CNBHD, BDC1, and PASCap. In that case, the free energy of these states will change upon Ca^{2+} -CaM binding. Consequently, the free energy difference between the states, and thereby the apparent voltage sensitivity of the transitions, will change too. Hence, we represented the effect of different degrees of Ca^{2+} -CaM binding as different voltage shifts in λ , κ_L , γ , as indicated in Figure 9E.

While the Markov model describes the time and voltage dependence of state occupancies, the experimental observable is current. From the model's occupancies, current was obtained by application of the GHK flux equation (equation 5) for both open states. While the single channel conductance of the two open states in the model is identical, the effective occupancy of O_2 and, thereby, its macroscopic conductance is drastically reduced by a flicker block, a rapid transition into C_2 , and a slower return to O_2 . To match experimental results, in particular the pre-pulse-dependent activation and tail current amplitudes, the flickering kinetics had to be adjusted to achieve a limiting occupancy ratio of 5.5 between C_2 and O_2 , i.e. $r_{\text{close}}/r_{\text{open}} = 4.5$. Typically, flicker-

A**B**

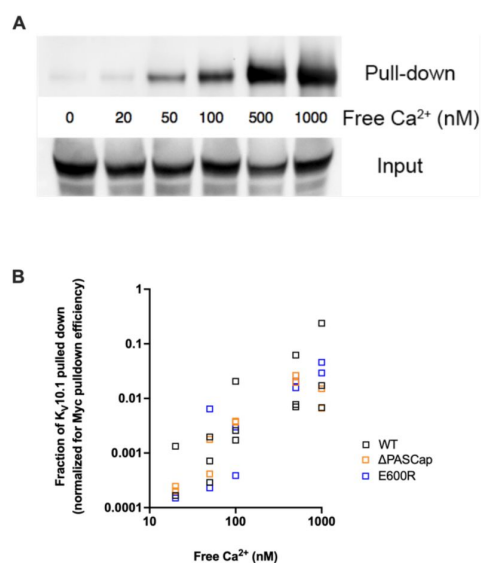
Supplement 2 to Figure 8.

The upper traces show the average of normalized ramps at the indicated times for Δ PASCap (A) and E600R (B) after induction of Ca^{2+} rise. At 60, 150 and 300 s, the dashed line corresponds to the ramp at time 0. The shadowed area indicates SEM (N: Δ PASCap = 8, E600R = 10). The lower panels correspond to the first derivative of the current traces to illustrate the changes in slope.

A**B**

Supplement 3 to Figure 8.

Like in Supplement 2, the upper traces show the average of normalized ramps for $\Delta\text{PASCap}^{\text{L322H}}$ (A) and $\text{E600R}^{\text{L322H}}$ (B) at the indicated times after induction of Ca^{2+} rise. At 60, 150 and 300 s, the dashed line corresponds to the ramp at time 0. The shaded area indicates SEM (N: $\Delta\text{PASCap}^{\text{L322H}}$ = 5, $\text{E600R}^{\text{L322H}}$ = 11). The lower panels correspond to the first derivative of the current traces to illustrate the changes in slope.



Supplement 4 to Figure 8.

A. Ca²⁺-dependent binding of CaM to Kv10.1. CaM was precipitated with anti-Myc antibody and the resulting pulled down fraction was immunoblotted using polyclonal anti-Kv10.1 antibody. B Quantification of all experiments reveals a strong dependence on Ca²⁺ concentration in WT and to a similar extent in mutant channels. The fraction of Kv10.1 pulled down increases by two orders of magnitude in high Ca²⁺.

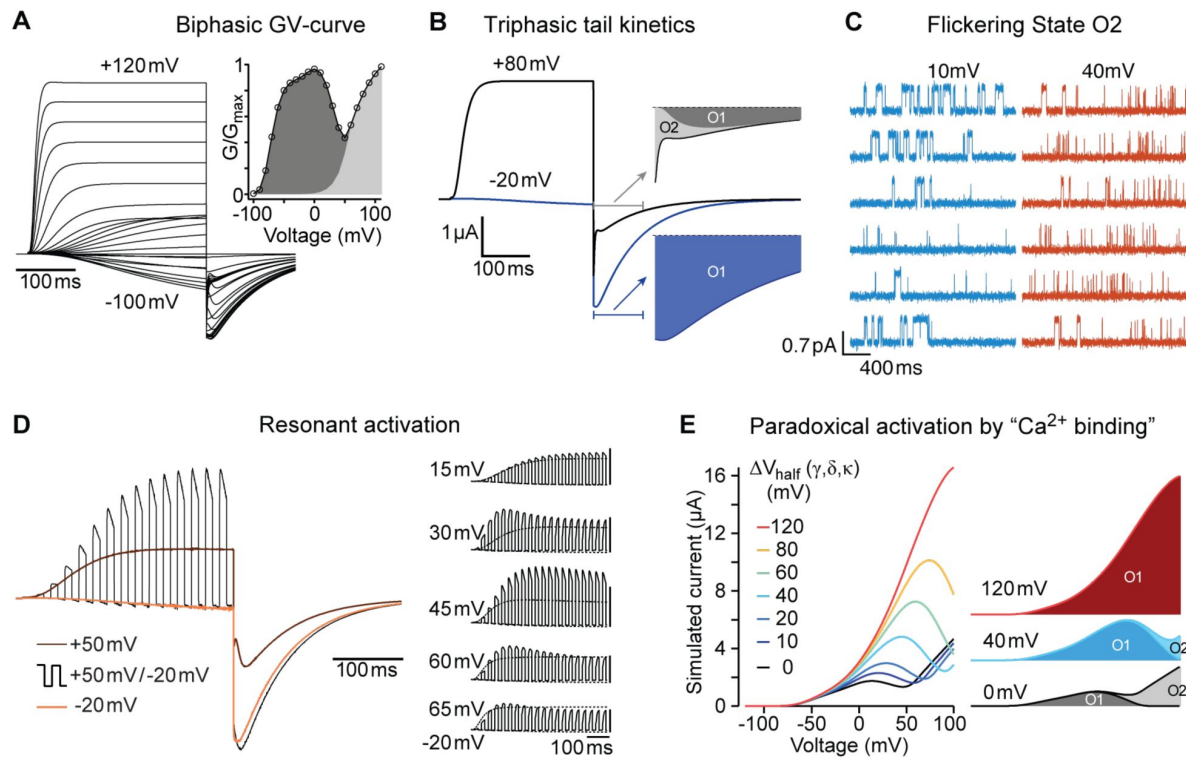


Figure 9.

A single model can reproduce all experimental observations.

A. In response to an I-V-protocol (**Fig. 1A**), the model displays biphasic activation, more clearly represented in the derived G-V curve (right, compare to **Fig. 1B**). The filled area indicates the contributions of the two open states. The tail-currents show a complex dependence on test-pulse voltage and time. **B.** Two traces from A, shown with longer repolarization. The first 100 ms of the tail currents are displayed on an extended timescale. Colored areas indicate the contributions of the two open states. After weak depolarization to -20 mV, the tail-current originates almost entirely from the mutant-specific open state O1 (blue area). After strong depolarization, O2 mediated currents initially dominate (light grey). For details, see Supplement 2 to **Figure 9**. **C.** The model predicts the differences in single channel behavior of Δ PASCap upon depolarization to +10 or +40 mV (Compare to **Figure 5**). **D.** In response to 10 ms pulses alternating between -20 and +50 mV, the model shows currents that exceed the currents obtained with constant pulses to +50 mV (compare **Fig. 4** B). Relating the currents during the positive and negative pulses to the concurrent currents elicited by the two constant pulses (brown and orange), the ratio lies by about 1.6 for the period starting 200 ms after the pulse onset. This excess current depends sensitively on the duration and voltage of the two pulse components. The right series of simulations displays the results for 15 ms pulses to -20 mV, alternating with 15 ms pulses to voltages from 15 mV to 65 mV. The corresponding responses to the constant pulses are displayed with thin dotted lines. To facilitate perception of the excess current, the five groups of traces are scaled individually, so that the peak amplitude of the dotted response elicited by the stronger depolarizations is displayed at equal size throughout. The vertical scalebars correspond to the same absolute current. From top to bottom, the excess current ratios at 200 ms changes are 1.05, 1.38, 1.74, 1.46, and 1.16. **E.** The binding of Ca^{2+} -CaM is implemented through change in the activation energy, corresponding to a shift in the equilibrium voltage of the gating transitions. The decomposition of the current into the individual open states' contribution shows that for increasing voltage shifts – representing high $[\text{Ca}^{2+}]_i$ – the mutant-specific O1 closes later into the ramp, until eventually all current is carried by O1.

Supplement 1 to Figure 9.

Cartoon depicting the state model proposed. A two-layer Markov-Model depicting possible conformations for the sensor and the ring. The sensor in each subunit can independently adopt one of three conformations (up, middle, and down) (Han et al., 2023). The up conformation represents the active conformation. The conformation of the ring changes across the two layers. The second conformation in the upper layer allows the channel to access “O2” provided that all sensors are in the up conformation. Transitions within each layer is governed by (α , β , γ , δ), we proposed that the transitions are identical in the two layers. Transitions between the two levels are governed by the rate constants κ and λ

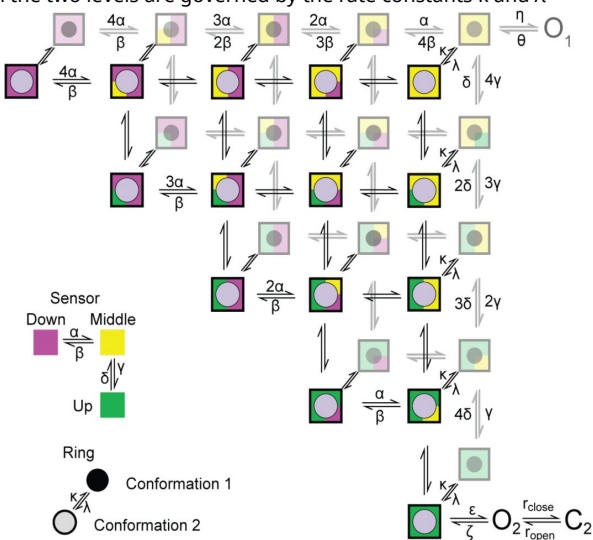
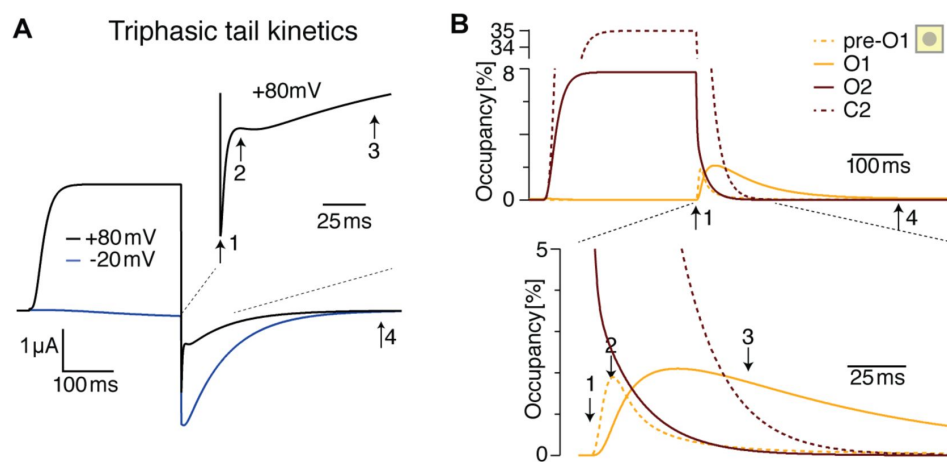


Table S2

parameters describing the voltage dependence of the transitions between states in the Markov model of KV10.1 Δ PASCap, KV10.1 E600R, and KV11.1 in conjunction with Eq. 6 and the structure in Supp. 1 to Fig. 9.

KV10.1 Δ PASCap					KV10.1 E600R					KV11.1				
rate	x_0	A_x	V_{hx}	K_x	x_0	A_x	V_{hx}	K_x		x_0	A_x	V_{hx}	K_x	
x	[s ⁻¹]	[s ⁻¹]	[V]	[V ⁻¹]		[s ⁻¹]	[V]	[V ⁻¹]			[s ⁻¹]	[V]	[V ⁻¹]	
α	8	192	0.09	-55	8	192	0.09	-55		0.5	99.5	0.09	-30	
β	4	596	-0.13	80	4	596	-0.13	80		1.2	16.8	-0.1	80	
γ	5	595	0.05	-30	5	595	0.05	-30		2.5	297.5	0.05	-30	
δ	15	235	0.055	30	15	235	0.055	30		7.5	117.5	0.025	30	
ϵ	200	0	—	—	200	0	—	—		200	0	—	—	
ζ	660	0	—	—	1440	0	—	—		24000	0	—	—	
η	80	320	0	-80	80	320	0	-80		80	320	0	-80	
θ	16	44	0	-50	16	44	0	-50		16	44	0	-50	
κ_R	0.125	69.875	-0.1	200	0.125	69.875	-0.1	200		0.65	119.35	-0.1	200	
κ_L	0.125	499.875	0.09	-130	0.125	499.875	0.09	-130		0.65	499.35	0.09	-130	
λ	0.1	999.9	-0.12	170	0.1	999.9	-0.12	170		0.1	999.9	-0.115	180	
r_{open}	100	0	—	—	30	0	—	—		30	0	—	—	
r_{close}	450	0	—	—	330	0	—	—		5970	0	—	—	



Supplement 2 to Figure 9.

The triphasic tail currents result from transient re-population of O1 during channel deactivation A. Simulated currents from Fig 9 B display simple, monophasic tail currents after weak depolarizations but triphasic tail current after +80 mV depolarization, similar to experimental results (see Fig. 1E). B. Occupancy of O2, O1 and its neighboring state as a function of time during a depolarization to +80 mV and repolarization to -100 mV. The three phases result from rapid initial vacuation of O2, followed by a slower process resulting from the delayed vacuation of C2 through O2. This results in an increased occupancy of the neighboring state to O1 (pre-O1), which drives a transient increase in O1 occupancy that is the basis for the transient tail current increase around time point 2. Slowly, deeper closed states are repopulated, and in the process, pre-O1 and eventually O1 fall empty from time point 3 to 4.

closures are thought to result from instabilities of the open state, unrelated to voltage sensor rearrangements. Here, r_{close} and r_{open} are constant and voltage-independent. The range of plausible absolute values for r_{close} and r_{open} is well constrained by experiments. If, following strong depolarizations, O_2 is populated faster than it is vacating into C_2 , a transient overshoot appears shortly after depolarization onset. This is observed in E600R, but not Δ PASCap (**Figure 1A** and **1C**). We found that the E600R observations can be mimicked with $r_{\text{open}}=30$, while mimicking Δ PASCap activation requires at least $r_{\text{open}}=100$. Further increases induce only small changes, visible in the triphasic tail current kinetics. In the limit of very fast flickering kinetics, the model becomes indistinguishable from a model without flickering in which O_2 microscopic conductance is reduced to 1/5.5.

To test the model, we focused on the most conspicuous kinetics features observed with Δ PASCap. The first feature was the tail kinetics. In contrast to the slow monophasic deactivation observed in response to *weak* depolarizing pulses (-20 mV), triphasic tail kinetics was detected in response to *strong* depolarizing pulses ($+80$ mV) (**Fig. 1C**). The model could replicate the slow deactivation after *weak* depolarizations, fast after *strong* depolarizations, and mixed kinetics on *moderate* stimuli (**Fig. 9B**; Supp. 2 to Fig. 9).

The model also reproduces the effect of a hyperpolarizing prepulse on different test potentials. As described (**Fig. 6**), O_1 is preferentially accessed from deep closed states, which correspond to states towards the left in the lower layer in the model. It is plausible that a hyperpolarizing prepulse (-160 mV) drives the channel to occupy the deep closed states (lower layer), while a prepulse of -100 mV distributes the channel between both layers. We, therefore, adjusted the parameters accordingly (**Table S2**) and simulated the current trace (Supp. 3 to Fig. 9). We focused on a test pulse that represents *moderate* depolarizations ($+20$ mV), and compared it to a *strong* depolarizing pulse ($+80$ mV). Like in the experiment, the more strongly hyperpolarizing prepulse (-160 mV) potentiates the current at $+20$ mV, as compared to the prepulse to -100 mV (Supplement 3 to **Fig. 9**, A). This does not happen for the larger depolarization to $+80$ mV (Supp. 3 to Fig. 9, B).

The model also recapitulates the behavior of Δ PASCap during repeated short stimuli between -20 and $+50$ mV. The current amplitude driven by intermittent stimuli is larger than that driven by a sustained stimulus to the same potential. The model also reproduces the relative size and the kinetics of the tail current (Fig. 9C.; Supp. 4 to Fig. 9).

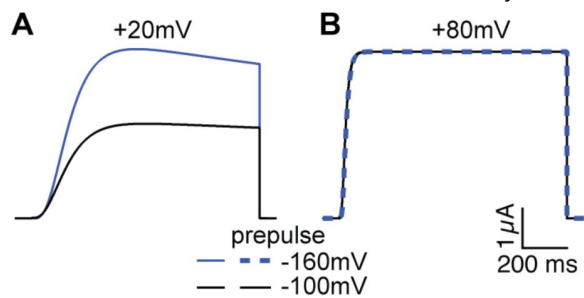
The effect of a rise in Ca^{2+} can be reproduced by shifting the voltage dependence of just two transitions ($\gamma \leftrightarrow \delta$ and κ). The rationale behind this is that Ca^{2+} -CaM binding stabilizes the basal configuration of the ring and thereby shifts the activation energy for state transitions that are coupled to ring reconfigurations, i.e. the second VSD displacements ($\gamma \leftrightarrow \delta$) and the ring gating itself (κ). Shifts in activation energy of voltage-dependent rates correspond to an extra term that can be expressed as voltage offset. When the voltage-dependence of these three rate constants is shifted by 10 to 120 mV, the occupancy of states in the “lower” layer is increased and thereby the access to O_1 (**Fig. 9D**). The resulting simulated currents reproduce the experimental observations: a linear current response during ramps in high intracellular Ca^{2+} .

To describe the behavior of $\text{Kv}10.1$ WT, we only needed to remove the access to O_1 and shift the parameters of the rotation of the ring in the hyperpolarizing direction to reflect the more stable structure resulting from intact interactions among intracellular domains and between these and the core of the channel. The model recapitulated the change in kinetics depending on the pre-pulse potential.

The model structure and the values of its parameters have been developed to reproduce the kinetic features of currents through $\text{Kv}10.1$ Δ PASCap channels. Can it provide insights into specific properties of other genotypes? A very palpable, E600R-specific feature is the transient current

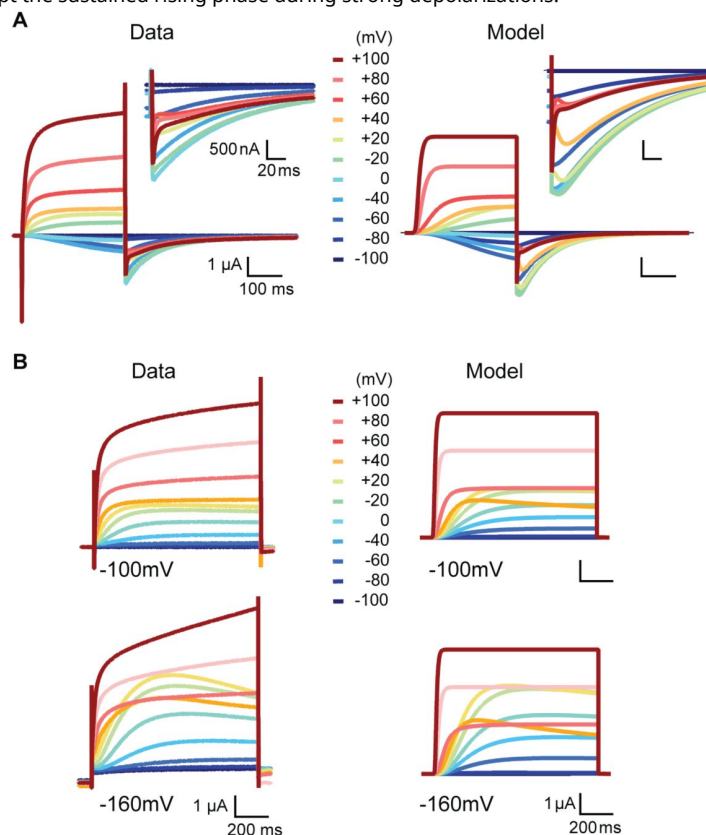
Supplement 3 to Figure 9.

Current simulations for Δ PASCap upon depolarization to +20 and +80 mV after a conditioning pulse to -160 or -100 mV. The current at +20 mV is larger after -160 mV, while the current at +80 mV is unaffected by the conditioning pulse (see Fig. 6A [↗](#))



Supplement 4 to Figure 9.

Side-by-side comparison of experimental data and model prediction during depolarizations between -100 and +100 mV in the presence of high (60 mM) (A) or low extracellular $[K^+]$ (2.5 mM) (B). The model describes the features of the experimentally observed currents, except the sustained rising phase during strong depolarizations.



peak that occurs within about 30 ms after depolarizations above +70 mV (**Fig 1A** and **1C**). It suggests that O_2 , the open state underlying current at these voltages, is populated and vacated with very similar rates. In the model, such a transient peak can indeed be reproduced if the rate constant from O_2 into C_2 (Supplement 1 to **Figure 9**) is reduced from 100 s^{-1} for ΔPASCap to 30 s^{-1} for E600R (Supp. 5 to Figure 9). E600R also supports larger tail currents, relative to the test-pulse currents, indicating an even stronger bias away from O_2 into C_2 . To replicate this, $r_{\text{close}}/r_{\text{open}}$ was increased to 11 and the rate constant ζ , leading out of O_2 was adjusted to compensate for the lower O_2 occupancy. These few changes, all concerned with O_2 instability, underly the simulations in **Supp. 5 A and B to Figure 9**. The corresponding parameters are given in **Table S2**.

The tails observed in E600R show a shape reminiscent of that of $K_v11.1$, where slow deactivation is crucial for its physiological function. Our model can mimic $K_v11.1$ if the flicker block is biased further towards the closed state, and the transitions between closed states are slowed (**Supp. 5 D and E to Figure 9** and **Table S2**),

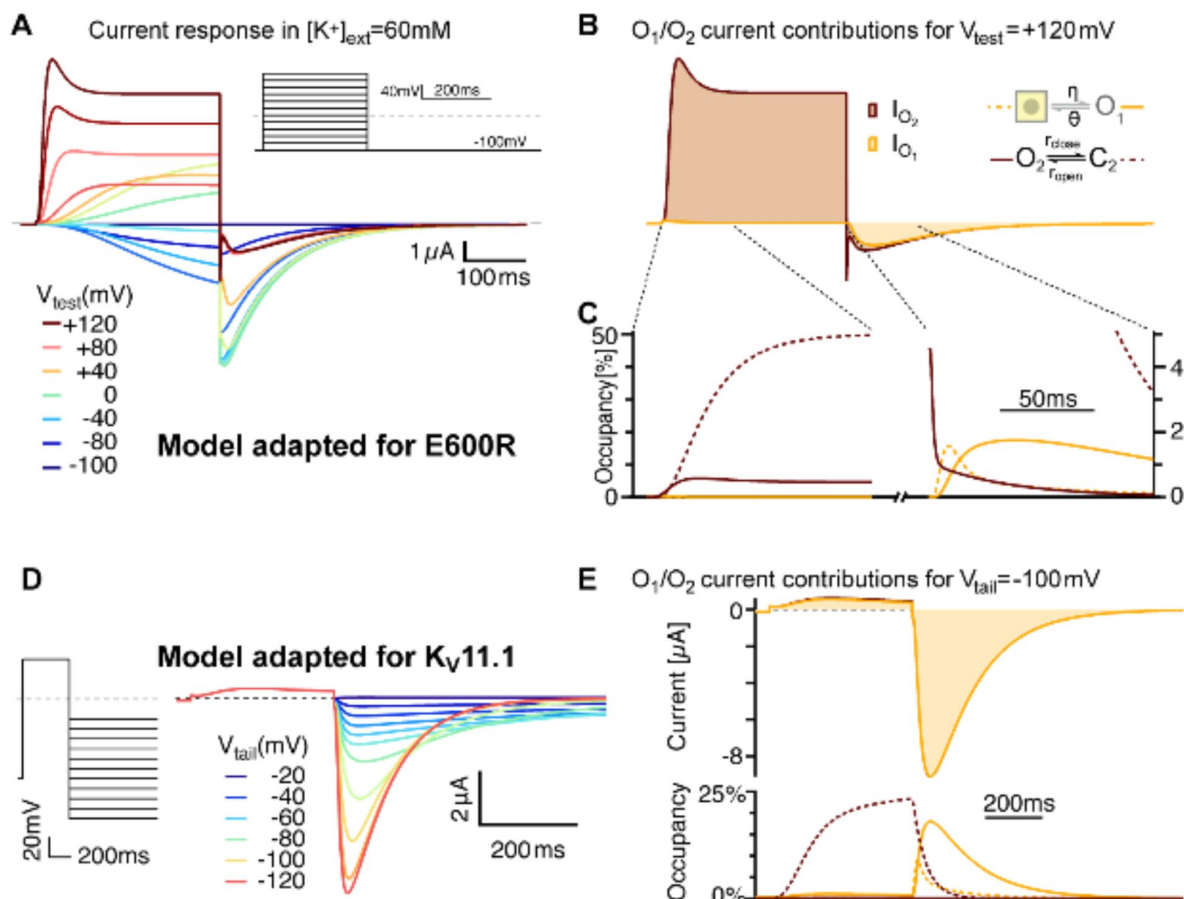
Discussion

The wide diversity of electrical responses in cells relies greatly on subtle differences in the behavior of voltage-gated channels. Despite the many relevant advances in the knowledge of the structure of ion channels, the correlation between the structures and the functional determinants of channel behavior is incompletely understood. In this study, we have combined biophysical, biochemical, and mathematical approaches to understand the complex gating behavior of $K_v10.1$ potassium channels, which is the basis of a group of diseases with devastating consequences (e.g. (Toplak et al. 2022; Tian et al. 2023)).

In *KCNH* channels, intracellular domains contribute to gating by forming an intracellular ring that rotates in response to depolarizing stimuli (Mandala and MacKinnon 2022). We have studied a series of mutant channels where the integrity of the intracellular ring is compromised by either deletions or point mutations. In all the mutants, the G-V relationship shows a biphasic behavior with evident inward rectification at intermediate depolarizations and a complex deactivation with kinetics also dependent on the stimulus potential (**Fig. 1**). Our observations can be explained by the coexistence of two different open states, one of which corresponds to the open state observed in WT (O_2), while the other one (O_1) is made visible as a result of the modification of the intracellular gating ring. We hypothesize that O_1 corresponds in the WT to the “non-conducting state 2” identified in the closely related $K_v10.2$ channel by cryo-EM (Zhang et al. 2023), in which flipping of Y460 (Y464 in $K_v10.1$) renders a hydrophobic constriction wider than 8 Å, enough to allow K^+ flow, but still corresponds to a non-conducting state. The presence of an intact intracellular ring would preclude ionic flow in the WT, and its alteration would explain the permeability of this state in the mutants.

The coexistence of two independent channel populations with different gating properties could be an alternative explanation for the observed behavior. Still, it is very unlikely that two such populations would be expressed at the same ratio in all oocytes, given the variability of the system, and the behavior of each of the mutants was highly reproducible among oocytes obtained from different frogs and time points. Furthermore, activity in excised patches from oocytes expressing mutant channels was abolished in the presence of astemizole.

The atypical open state O_1 allows the study of intermediate steps between the two major gating events, VSD displacement and ring rotation. Upon intermediate depolarization, the channels would access the open states sequentially, while O_2 is predominant on strong depolarizing steps and O_1 upon mild stimuli. The analysis of the GV of the mutants using global parameters (**Fig. 2**) revealed that they all share the component responding to mild depolarizations (O_1) and only the



Supplement 5 to Figure 9

Model variants can account for features of related channels and mutated variants. A E600R gating features are consistent with slower flickering of a less populated O_2 . After lowering the stability of O_2 (larger r_{close}/r_{open} and larger ζ/ϵ) and an overall slower flickering between O_2 and C_2 (Table S2), simulated currents show transient peaks during strong depolarizations and larger tails with more pronounced triphasic waveforms. Those features resemble the measurements with the E600R mutant (Fig 1A). B The relative currents of C_2 and O_2 differ even more than in the ΔPASCAP model. The transient peak results from a delayed O_2 - C_2 equilibration due to reduced flicker rates. Traces are shown in the same units as in A. C As close-up of the states' occupancies shows that the mechanism of triphasic tail currents remains the same in the E600R model version (compare Supplement 2 to Figure 9). D The model recapitulates key features of Kv11.1 when the balance between O_2 and C_2 is further biased towards the closed state and with slowed transitions between closed states (see Table S2). The traces were generated for high extracellular $[K^+]$ (50 mM), to allow for comparison with experimental recordings in (Lörinczi et al. 2015).

second component (O_2) and (probably most importantly) the transition between components depends on the particular mutant studied. The larger the deletion in the intracellular ring is, the stronger the shift in the voltage dependence of the second component.

Importantly, this observation is also incompatible with two independent populations of channels. Since the two gating steps are sequential (Mandala and MacKinnon 2022 [↗](#)), the displacement of the VSD remains the main factor governing the speed and voltage dependence of the activation. Thus, changes in the extracellular Mg^{2+} concentration, which are known to interfere with the movement of the VSD (Bannister et al. 2005 [↗](#); Silverman et al. 2004 [↗](#)), cause a shift of the voltage dependence and activation speed that affects both gating components (Fig. 3 [↗](#)), but not the transition between them.

The biphasic behavior arises from a different apparent conductance between the two open states, with larger open probability and, therefore, larger macroscopic amplitude for O_1 . This results in a decrease in current amplitude as the channels leave O_1 and transition to O_2 . Since the time required to access the two states is different, when short alternating stimuli are applied, each with a too short duration to allow entry into O_2 , the longer open time of this state results in a larger current amplitude than when a single sustained stimulus is used (Fig. 4 [↗](#)). The presence of a second state with larger or longer openings when the integrity of the ring is compromised could explain the larger current amplitudes observed in heteromeric $K_{v11.1}$ (HERG1a/1b) channels, which lack at least one PAS domain as a result of alternative splicing and are crucial for proper cardiac repolarization (Feng et al. 2021 [↗](#)). Thus, the second open state could also have physiological relevance in naturally occurring channel complexes.

Access to O_1 is favored from deep deactivated states, which appear important in other aspects of $K_{v10.1}$ channel function (e.g., Cole-Moore shift). Our conclusion is based on the changes in amplitude observed with different pre-pulse potentials (that is, driving the population to deep closed states, Fig. 7A and B [↗](#)) and on the behavior of a mutant known to hinder access to such deactivated states (L322H), which, when combined with mutations revealing O_1 , shows limited access to this state (Fig. 7C and D [↗](#)).

Ca^{2+} -CaM is an important modulator of $K_{v10.1}$ that reduces WT current amplitude in the presence of elevated Ca^{2+} levels. A paradoxical current increase had been described for some of the mutants used in this study (Lörinczi et al. 2016 [↗](#)), and we speculated that the presence of O_1 could be the basis for this phenomenon. Indeed, the elevation of intracellular Ca^{2+} leads to a transient loss of the biphasic behavior and a larger current amplitude of the mutants compatible with a larger fraction of channels in O_1 . Because access to O_1 occurs from deep closed states, this could be explained by an increased occupancy of such deactivated states in response to CaM binding. This appears to be the case since CaM induces a biphasic behavior in the mutant channels that show reduced access to deep closed states; thus, L322H mutants behave like the parental variants in the presence of Ca^{2+} -CaM. This implies a mechanistic explanation for the effect of Ca^{2+} -CaM on WT since favoring entry into deep closed states would result in a decrease in current amplitude in the absence of (a permeable) O_1 .

Our initial hypothesis that CaM participates constitutively in the gating machinery of the channel, based on the loss of biphasic behavior when the C-terminal binding site for CaM was mutated (Fig. 8D [↗](#)), is unlikely to be correct because although there is a significant binding of CaM to the channel at basal intracellular Ca^{2+} , this fraction of channels, combined with the strong increase in bound CaM in the presence of high Ca^{2+} and the variable intracellular basal Ca^{2+} in oocytes would be insufficient to explain the qualitatively consistent behavior of the current of the different mutants. We speculate that the effects of the mutations in CaM binding sites are more related to their location in the protein than to their ability to bind CaM.

In summary, the gating of $K_V10.1$ (and similar channels) consists of a sequence of events that affect the voltage sensing domain, which moves in two sequential steps (Schönherr et al. 1999 [\[1\]](#)) and whose movement is transferred to the pore domain through intramolecular interactions (Bassetto et al. 2023 [\[2\]](#)). The voltage sensor maintains the gate closed (Mandala and MacKinnon 2022 [\[3\]](#)), and its displacement has a permissive effect on the gate opening. Once this restrictive factor is removed, a second step occurs, most likely corresponding to a rotation of the intracellular ring that finally allows the gate to relax to the open state (Patlak 1999 [\[4\]](#)). This final step is the only one directly observed in WT channels.

We proposed the existence of a second open state in $K_V10.1$ with an altered eag domain, because that could readily explain otherwise puzzling observations in macroscopic recordings. Our single channel recordings provided definitive evidence for channel openings with two different kinetic characteristics. Finally, we constructed a Markov model of $K_V10.1$ mutants' gating that replicates all key experimental observations (**Fig. 9 and its [\[5\]](#) supplements**). This model performance required two modifications to previously established K_V10 models: 1. introduction of an additional open state with a larger macroscopic current and 2. an additional gating step orthogonal to the previously described state plane. This added a second layer of states. Steps within each layer correspond most likely to two sequential displacements of the four VSD. Steps between the layers are interpreted as reconfiguration of the intracellular ring. The new state O_1 is already accessible when the first displacement of all four VSD is completed but the ring has not changed, while entry into O_2 requires complete VSD displacements and ring reconfiguration. The rapidly flickering open state of O_2 observed in the single channel recordings, is realized in the model by rapid, voltage-independent gating in-and-out of a neighboring closed state (C_2). This flicker block might also offer an explanation for a feature of the mutant channels, that is not explained in the current model version: the continued increase in current amplitude, hundreds of milliseconds into a strong depolarization (Supp. 4 to Fig. 9). If the relative stability of O_2 and C_2 continued to change throughout depolarization, such a current creep-up could be reproduced. However, this would require either the introduction of further layers of $O_n \leftrightarrow C_n$ states, or a non-Markovian modification of the model's time evolution.

To which degree does the model reflect the topology and gating transitions of $K_V10.1$ mutants? By construction, the presented model has one of the simplest structures that can replicate all key findings. Specifically, a model with less than three different gating steps, (two VSD, one ring), without flicker block, or with O_1 and O_2 in close proximity would fail to replicate at least one of the key findings. With respect to these fundamental properties, the model's structure should mirror the fundamental properties of $K_V10.1$ gating. The finer details of the model structure are less certain; we were able to vary them without large improvements in performance, but also without clear losses. This includes the feature that trans-layer connections exist for each state except the open states and the flicker block, and the feature of complete equivalence of the VSD transitions within lower and upper layer. Those features were made as choices for simplicity and symmetry. One price for the simple structure is the relative complex voltage dependence of the rates. The functional form of the rates' voltage dependence is an often-underappreciated way to shift model complexity from the state diagram to the rate functions. Following Arrhenius or Eyring formalisms, transition rates are expected have the form $c_0 \cdot \exp(\Delta V/k)$ and range from zero to infinity without saturation or a non-zero minimal rate. If, however, a transition between two states ($\leftrightarrow S_1 \leftrightarrow S_2 \leftrightarrow$) is used as a shorthand for an underlying linear three-state system ($\leftrightarrow S_1 \leftrightarrow S_2 \leftrightarrow S_3 \leftrightarrow$), in which only the transition $S_1 \leftrightarrow S_2$ is voltage-dependent, while $S_2 \leftrightarrow S_3$ features constant forward and backward rates, then the current activation kinetics in the two-state shorthand will saturate even for infinitely high voltages. Similarly, a non-zero basal rate x_0 can be caused by an alternate "leakage" route from the first state S_1 to the open state O that occurs at a very low, voltage-independent rate. In the light of such considerations, the sigmoidal voltage dependences of the model's rates point to a much more complicated underlying structure.

A large majority of ion channel models in use today uses voltage dependent rates with functional forms that are more complex than a simple exponential. Although in general, we prefer the use of the simple forms (e.g. (Yang et al., 2016 [DOI](#))), in the present case, we decided to keep the model structure as simple as possible and use a more complex voltage dependence of the rates, to avoid adding even more states. As a consequence, some voltage dependencies can raise questions about how the model corresponds mechanistically to underlying gating steps. In particular rate κ , governing the transition from the newly introduced upper floor to the ground floor, is described by a sum of two voltage dependencies of opposite sign. The necessity for such a complex voltage dependence clearly signals unresolved structural details. What's more, the κ/λ transition could reasonably be expected to be voltage independent because we related it to ring reconfiguration, a process that should occur as a consequence of a prior VSD transition. We have made some attempts to treat this transition as voltage independent but state-specific with upper-layer bias for states on the right and lower-layer bias for states on the left. This is in principle possible, as can already be gleaned from the similar voltage ranges of the left-right transition (α/β) and the κ_L/λ transition. However, this approach leads to a much larger number of free, less well constrained kinetic parameters and drastically complicated the parameter search. The model version presented here is therefore a low-complexity compromise that showcases the explanatory power of a second open state and an additional gating step. In summary, this study presents a more complete description of the gating mechanism of $K_v10.1$ channel, which, importantly, can be extended to other members of the *KCNH* family. In response to depolarization, the movement of the voltage sensor would have a permissive role in the opening of the gate. The rotation of the intracellular ring would be the effective unlocking mechanism allowing permeation. This has profound implications pertaining to the possibilities of fine-tuning gating since the intracellular ring is more susceptible to posttranslational modifications and protein-protein interactions than the transmembrane domains. Our current knowledge of the physiology and pathophysiology of $K_v10.1$ indicates that the channel is relevant for the regulation of excitability acting at potentials close to the resting rather than during active electrical signaling. Therefore, sustained modulation of gating, possible through modification of the intracellular ring, would be crucial for channel function. Since they allow dissection of the ring-dependent effect, our mutants will allow for a direct study of such modulation mechanisms.

Materials and Methods

Constructs

Mutants were generated using $K_v10.1$ (hEAG1) cloned in pSGEM (M. Hollmann, Bochum University) as a template (Jenke et al. 2003 [DOI](#)). The deletion mutants, $\Delta 2-10$ and ΔPASCap , were generated using In-Fusion HD Cloning kit (Clontech (TaKaRa)) following the supplier's instructions. For each construct, we designed two primers, each of them with two regions: a 3' region that anneals to the template immediately up- or downstream of the sequence to be deleted and a 5' that does not bind to the template but overlaps with the second primer (**Table S3** [DOI](#)). The subsequent PCR amplification will then omit the sequence between the hybridization sites for the primers.

To generate the point mutations ($\Delta \text{PASCap}^{\text{L322H}}$, $\Delta \text{PASCap}^{\text{BDN}}$, $\Delta \text{PASCap}^{\text{BDC2}}$, $\text{E600R}^{\text{L322H}}$, $\text{E600R}^{\text{BDN}}$, $\text{E600R}^{\text{BDC2}}$), site-directed mutagenesis was performed using Quick Change II XL kit (Agilent Technologies) using the primers listed in **Table S3** [DOI](#). E600R.pLeics71 and $\Delta \text{eag.pLeics71}$ were a gift from Dr. John Mitcheson, University of Leicester.

The expression construct for 5-myc-calmodulin was obtained by restriction cloning of CALM1 from pKK233-hCaM, which was a gift from Emanuel Strehler (Addgene plasmid # 47598) (Rhyner et al. 1992 [DOI](#)) into a pSGEM construct with five consecutive repeats of the myc tag (Lörinczi et al. 2015 [DOI](#)) using NcoI and HindIII (New England Biolabs).

All plasmids were linearized with NheI, and cRNA was synthesized using mMESSAGE mMACHINE T7 Transcription kit (Invitrogen Ambion).

Two-Electrode Voltage-Clamp Recordings

Oocyte preparation and injection were performed as described (Stühmer 1992). The amount of cRNA injected depended on the current amplitude obtained with each construct: 0.075–0.5 ng/oocyte (WT and point mutation), 0.075–5 ng/oocyte (deletion mutants), and 8–10 ng/oocyte (split channels). Oocytes were maintained at 18°C in ND96 buffer (in mM: 96 NaCl, 2 KCl, 1.8 CaCl₂, 1 MgCl₂, 5 HEPES, 2.5 Na-pyruvate, 0.5 Theophylline, pH 7.55). Tetracycline (USB) (50 µg/ml), Amikacin (Enzo) (100 µg/ml), and Ciprofloxacin (Enzo) (100 µg/ml) were added as recommended in (O'Connell et al. 2011 [↗](#)).

Two-electrode voltage-clamp (TEVC) recordings were performed 1–5 days after oocyte injection. The intracellular electrodes had a resistance of 0.4–1.5 MΩ when filled with 2 M KCl. Normal Frog Ringer solution (NFR): (in mM: 115 NaCl, 2.5 KCl, 10 HEPES, 1.8 CaCl₂, pH 7.2), was used as external solution. Higher K⁺ concentration (mM: 60 KCl, 57.5 NaCl, 10 HEPES, 1.8 CaCl₂, pH 7.4), was used instead to examine tail currents. Cl⁻-free NFR (in mM: 115 Na-methanesulfonate, 2.5 KOH, 10 HEPES, 1.8 Ca(OH)₂, pH 7.2), was used to limit current contamination with Cl⁻ current; in this case, agar bridges (2% agar in 3 M NaCl) were used for the reference electrodes. To raise the intracellular Ca²⁺ concentration (Lörinczi et al. 2016 [↗](#)), 5 µM ionomycin (Abcam) and 5 µM Thapsigargin (Abcam) were added to the bath. Ionomycin and Thapsigargin 5 mM stocks were prepared using DMSO and diluted in the recording medium (Cl⁻-free NFR) immediately before recording. The final concentration of DMSO was thus 0.1%.

Data acquisition was performed using a TurboTEC 10-CD amplifier (npi Electronics) and the ITC-16 interface of an EPC9 patch-clamp amplifier (HEKA Elektronik). The current was filtered at 1.3 KHz and sampled at 10 KHz. Patchmaster software (HEKA Elektronik) was used to design and apply the stimulus protocols applied. Because of the profound effects of hyperpolarization on channel kinetics, leak subtraction was avoided except when explicitly indicated. Linear leak was subtracted offline assuming a reversal potential of 0 mV. Fitmaster (HEKA Elektronik) and IgorPro (WaveMetrics) were then used to analyze the recordings.

Most conductance-voltage plots were obtained for recordings with [K⁺]_{ext} = 60 mM. In this condition, tail currents are large and allow precise estimation of the reversal potential V_{eq} . Under this condition, the difference between intra- and extracellular potassium concentration is small and the Goldman-Hodgkin-Katz flux equation predicts a nearly linear relation. In these cases, overall conductance was calculated from the end-pulse current after measuring the reversal potential (V_{eq}) using:

$$G = I / (V_m - V_{eq}) \quad \text{Eq.2}$$

where I is the current amplitude and V_m the stimulus. Reversal potential was calculated using the end pulse current for mutant channels activating below the equilibrium potential for potassium, or instantaneous IV using tail currents for channels opening at more depolarized potentials. Conductance was then normalized to the maximum value and plotted against voltage stimulus. For WT recordings (**Fig. 1B** [↗](#)), a sigmoidal response was then fitted with a Boltzmann equation:

$$f(V) = \left(1 + \exp \left((V_h - V_m) / K \right) \right)^{-1} \quad \text{Eq.3}$$

With V_h being the voltage for half-maximal activation, K the slope factor and V_m the membrane potential as above.

Deletion/Mutation	Template (s)	Primers
Δ2-10	pSGEM.Kv10.1 pSGEM Kv10.1 L341Split	F attcgatatcaagcttatgggtggccctcaaaacacgtttct R cggtatcgataagcttcagctggctccaaaatgtctctct
ΔPASCap 2-25	pSGEM Kv10.1	F gctgccgccaccatg aatgataactaattttgtgttggggaa R catgggtggcggcagctcg
ΔCNBHD 525-697	psGEM Kv10.1.	F tccagaggcattgac aatgaggccccctgatcttgc R gtcaatgcctctggacatggaccaa
L322H	pSGEM Kv10.1 ΔPASCap pSGEM Kv10.1 E600R	F gatgagggcatcagcagccattcagctctctaaaagtgt R acaacttttagagagctgaaatggctgctgatgccctcatc
N-terminal CaM binding site (BDN) F151NL154N	pSGEM Kv10.1 Δ2-10 pSGEM Kv10.1 ΔPASCap	F151N ctcttgtcagccgagcattcttccccagcctttac L154N tgcttgtcagtgctcttctgtattccgagcattcttccccag
C-terminal CaM binding site (BDC2) F714SF717S	pSGEM Kv10.1 Δ2-10 pSGEM Kv10.1 ΔPASCap	F714S gtcggaatctctggctgaggcgccggacag F717S ctgctgtcggcttctctggctgaggcgccgg

Table S3.

Primers used for infusion cloning or site-directed mutagenesis.

The sequences are listed 5'-3'. For mutagenesis primers, only the sense sequences are given. The reverse primers corresponded to the reverse-complement sequence.

The biphasic response we observed in the mutants was described with two sigmoidal components and a weight W to represent the transition between the two components. The equation used was as follows:

$$f(V) = A_o + (1 - W(V)) \cdot A_1 \cdot (1 + e^{(V_{h1} - V_m)/K_1})^{-1} + W(V) \cdot A_2 \cdot (1 + e^{(V_{h2} - V_m)/K_2})^{-1}$$

$$W(V) = (1 + e^{(V_{h3} - V_m)/K_3})^{-1} \quad \text{Eq.4}$$

If the currents were recorded in $[K^+]_{\text{ext}}=2.5$ mM, the driving force for currents becomes considerably non-linear (Kotler et al. 2022 [\[18\]](#)) and we estimated the conductance based on the full Goldman-Hodgkin-Katz flux equation for a current surface density Φ .

$$\Phi = P \cdot P^{\text{open}} \cdot zF \cdot \frac{V_m}{V_T} \left(\frac{[K^+]_{\text{in}} - [K^+]_{\text{out}} \cdot e^{-V_m/V_T}}{1 - e^{-V_m/V_T}} \right), \text{ where } V_T = \frac{RT}{zF}$$

, with the gas constant R , the Faraday constant F , the absolute temperature T , the potassium channel open probability P^{open} , and the maximum permeability P . The ion's valence z equals 1. Because we only work with normalized conductances, the following relation can be used to determine the voltage-dependent conductance:

$$G(V) \propto I \cdot \left(\frac{1 - e^{-V_m/V_T}}{[K^+]_{\text{in}} - [K^+]_{\text{out}} e^{-V_m/V_T}} \right) \cdot \frac{V_T}{zFV_m} \quad \text{Eq.5}$$

Single-channel recording and analysis

The vitelline membrane of oocytes expressing Δ PASCap cRNA, as described for two-electrode experiments, was mechanically removed after osmotic shock (Stühmer 1992) to expose the plasma membrane. Patch pipettes were pulled from thick-wall (0.5 mm) borosilicate capillaries to obtain a resistance of 7-12 M Ω when filled with the intracellular solution, which contained (mM) 100 KCl, 10 EGTA, 10 HEPES pH 7.2 (KOH). Recordings on outside-out patches were acquired using an EPC10 Plus amplifier and PatchMaster software (HEKA Elektronik). Currents were digitized at 10 kHz and filtered at 2 kHz. Analysis was performed using FitMaster. For presentation, the traces were digitally filtered at 100 Hz.

Pull-down

Xenopus laevis oocytes were co-injected with RNA coding for 5xmyc-calmodulin (10 ng) and K_v10.1 (0.5 ng). Oocytes were lysed 72 h after injection through mechanical disruption in lysis buffer and incubation for 30 min on ice. The lysis buffer (1% Triton X-100, 150 mM NaCl, 50 mM HEPES pH 7.4, cOmplete EDTA-free protease inhibitor cocktail (Roche)) contained different free Ca²⁺ concentrations (0, 20, 50, 100, 500, and 1000 nM). To obtain accurate Ca²⁺ concentrations, EGTA was titrated with CaCal₂, to prepare 100 mM CaEGTA stock solution as described in (Tsien and Pozzan 1989 [\[19\]](#)). CaEGTA and EGTA were then added to the lysis buffer, adjusting EGTA/CaEGTA to control the concentration of free Ca²⁺ taking into account pH, ionic strength and temperature of solution using MaxChelator (Bers et al. 2010 [\[20\]](#)). As controls, non-injected oocytes, and oocytes injected with only K_v10.1 or 5xmyc-calmodulin were lysed in a Ca²⁺ free buffer.

The lysate was then centrifuged (20,000 xg) at 4 °C for 3 min to remove debris. 10% of the supernatant was set aside to load on the gel as input control. The lysate was pre-cleared using protein G magnetic beads (New England Biolabs). The cleared supernatant was then incubated with 3 μ g anti-myc antibody (SIGMA M4439 monoclonal anti c-myc or Abcam ab206486 rat mAb to

myc tag (9E10)) for 1.5h at 4 °C. Protein G magnetic beads were then added and incubated for 1.5h at 4 °C in rotation. After magnetic retrieval, the beads were washed three times using 0.1% Triton X-100, 300 mM NaCl, 50 mM HEPES pH 7.4, cOmplete (EDTA-free protease inhibitor cocktail), EGTA and CaEGTA to obtain the corresponding Ca^{2+} -free concentrations (see above). Electrophoresis and immunoblotting conditions were as previously described (Lörinczi et al. 2015 [\[1\]](#)) using an anti-Myc (Sigma, 1:1000) or an anti-K_V10.1 antibody (Chen et al. 2011 [\[2\]](#)) overnight.

Markov-model construction

The standard model for K_V10.1 gating comprises two gating steps for each of the four subunits' voltage sensors. The first step unfolds in the hyperpolarized voltage range and forms the basis of the Cole-Moore shift, characteristic of K_V10.1 channel gating. The second, faster step readies the channel for opening, and once it is performed in all subunits, a conducting state can be reached (see also (Mandala and MacKinnon, 2022 [\[3\]](#))). While this model captures the key features of WT activation, we had to extend it to account for the features of mutant gating. Throughout the model development and tuning, we focused on the experiments performed with the ΔPASCap mutant. However, a limited number of tests with other parameters showed that experimental findings with E600R and even with K_V11.1 can also be matched (Supp.5 to Fig.9).

In a first step, we attempted a minimal model extension by attaching an additional open state to some closed state of the standard model. None of the resulting models, for any choice of attachment site and rate voltage dependence, could replicate the multi-phasic tail currents, the pre-pulse dependent activation, and the delayed opening. We came to this conclusion in two steps. First, we could exclude many possible attachment sites because the attachment site must be sufficiently far away from the conventional open state. Otherwise, the transition from “O1 preferred” to “O2 preferred” via a few closed intermediate states is very gradual and never produces the biphasic GV curves. Second, we found that the first gating transition in the standard model (along α , left to right) can either produce a sigmoidal current onset or bias the model for occupancy of state O₁ over O₂, but not both. Waveforms such as the dark orange trace in **Figure 6** [\[4\]](#) A, in response to a -160 mV pre-pulse and a 20 mV test pulse require both a bias towards O1 and a sigmoidal onset. We found that this can only be accounted for by introducing a third gating step orthogonal to the two transitions in the standard model.

Without *a priori* information about the new states introduced in this way, we decided to simply add a copy of the standard model as an additional layer, an “upper floor”. Crossing from the ground floor to the upper floor was made possible for any state. From a structural view, this newly added transition might be related to a reconfiguration of the gating ring. Considerations as given above led us to attach the mutant-specific O1 to the basal level and the conventional O2 to the upper level. The experimental data indicated that O1 had to be accessible to states “on the right of the ground floor”, that is, accessible from deep closed states. Along the same lines, O1 could not be located “too far toward the bottom” of the scheme, because it is an open state. Eventually, we decided to attach O₁ only to the state that corresponds to the completion of the first gating step with no further transitions (neither the second voltage-sensor transition nor the gating ring transition) in all four voltage sensors. In contrast, to enter O2, all subunits must complete both voltage sensor transitions and the collective gating ring transition. This model's structure is displayed in Supp. 1 to Fig.9.

We also tested models in which the upper floor was only accessible from a subset of states and models with O1 attached to more than a single closed state or even multiple O1 states with varying conductance. While those more complex models offered a gradual improvement matching experimental traces, they showed no striking advantages over the more symmetric and parsimonious model presented here. Furthermore, we have extensively tested variants with a broken symmetry in the gating kinetics. In these models, the conventional gating steps (rates α , β , γ , δ) differed between the ground floor and the newly introduced top floor states, e.g., by the

introduction of factors to the α , β , γ , δ values. Under these conditions, a local balance was achieved by corresponding inverse factors to the local κ and λ . Again, the gradual improvements that could be achieved did not seem to justify the introduction of the additional parameters.

Voltage dependence of rates

The simplest model structure that allowed us to reproduce the key observations (Supp. 1 to Fig.9) was implemented in Igor Pro as a system of coupled differential equations that govern the mean-field occupancies of the 33 states. To simulate the stochastic gating of a single channel, the Gillespie Algorithm was implemented (Gillespie 1976 [link](#)). For the voltage-dependent transition rates, we chose a sigmoidal functional form (Eq.6).

$$x = x_0 + \frac{A_x}{1 + \exp((V_m - V_x^h) \cdot k_x)} \quad \text{Eq. 6}$$

where x stands for one of the rates α , β , γ , δ , that govern transitions in the first and second dimension of the model, which we attribute to two sequential steps of voltage sensor displacement. The rates η and θ of opening and closing of the mutant-specific open state O_1 also follow this functional form, as does the rate λ , governing the return from the upper layer. The rate κ of entering the upper layer is the sum of two components, $\kappa_{JKK} = \kappa_K + \kappa_L$, which again follow the same functional form. The rates leading towards and away from the second open state O_2 , ε , ζ , r_{open} and r_{close} are constants.

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Data availability

The authors declare that the data supporting the findings of this study are available within the paper and its supplementary information files.

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Reviewer #1 (Public Review):

Gating of Kv10 channels is unique because it involves coupling between non-domain swapped voltage sensing domains, a domain-swapped cytoplasmic ring assembly formed by the N- and C-termini, and the pore domain. Recent structural data suggests that activation of the voltage sensing domain relieves a steric hindrance to pore opening, but the contribution

of the cytoplasmic domain to gating is still not well understood. This aspect is of particular importance because proteins like calmodulin interact with the cytoplasmic domain to regulate channel activity. The effects of calmodulin (CaM) in WT and mutant channels with disrupted cytoplasmic gating ring assemblies are contradictory, resulting in inhibition or activation, respectively. The underlying mechanism for these discrepancies is not understood. In the present manuscript, Reham Abdelaziz and collaborators use electrophysiology, biochemistry and mathematical modeling to describe how mutations and deletions that disrupt inter-subunit interactions at the cytoplasmic gating ring assembly affect Kv10.1 channel gating and modulation by CaM. In the revised manuscript, additional information is provided to allow readers to identify within the Kv10.1 channel structure the location of E600R, one of the key channel mutants analyzed in this study. However, the mechanistic role of the cytoplasmic domains that this study focuses on, as well as the location of the Δ PASCap deletion and other perturbations investigated in the study remain difficult to visualize without additional graphical information. This can make it challenging for readers to connect the findings presented in the study with a structural mechanism of channel function.

The authors focused mainly on two structural perturbations that disrupt interactions within the cytoplasmic domain, the E600R mutant and the Δ PASCap deletion. By expressing mutants in oocytes and recording currents using Two Electrode Voltage-Clamp (TEV), it is found that both Δ PASCap and E600R mutants have biphasic conductance-voltage (G-V) relations and exhibit activation and deactivation kinetics with multiple voltage-dependent components. Importantly, the mutant-specific component in the G-V relations is observed at negative voltages where WT channels remain closed. The authors argue that the biphasic behavior in the G-V relations is unlikely to result from two different populations of channels in the oocytes, because they found that the relative amplitude between the two components in the G-V relations was highly reproducible across individual oocytes that otherwise tend to show high variability in expression levels. Instead, the G-V relations for all mutant channels could be well described by an equation that considers two open states O1 and O2, and a transition between them; O1 appeared to be unaffected by any of the structural manipulations tested (i.e. E600R, Δ PASCap, and other deletions) whereas the parameters for O2 and the transition between the two open states were different between constructs. The O1 state is not observed in WT channels and is hypothesized to be associated with voltage sensor activation. O2 represents the open state that is normally observed in WT channels and is speculated to be associated with conformational changes within the cytoplasmic gating ring that follow voltage sensor activation, which could explain why the mutations and deletions disrupting cytoplasmic interactions affect primarily O2.

Severing the covalent link between the voltage sensor and pore reduced O1 occupancy in one of the deletion constructs. Although this observation is consistent with the hypothesis that voltage-sensor activation drives entry into O1, this result is not conclusive. Structural as well as functional data has established that the coupling of the voltage sensor and pore does not entirely rely on the S4-S5 covalent linker between the sensor and the pore, and thus the severed construct could still retain coupling through other mechanisms, which is consistent with the prominent voltage dependence that is observed. If both states O1 and O2 require voltage sensor activation, it is unclear why the severed construct would affect state O1 primarily, as suggested in the manuscript, as opposed to decreasing occupancy of both open states. In line with this argument, the presence of Mg²⁺ in the extracellular solution affected both O1 and O2. This finding suggests that entry into both O1 and O2 requires voltage-sensor activation because Mg²⁺ ions are known to stabilize the voltage sensor in its most deactivated conformations.

Activation towards and closure from O1 is slow, whereas channels close rapidly from O2. A rapid alternating pulse protocol was used to take advantage of the difference in activation and deactivation kinetics between the two open components in the mutants and thus drive an increasing number of channels towards state O1. Currents activated by the alternating

protocol reached larger amplitudes than those elicited by a long depolarization to the same voltage. This finding is interpreted as an indication that O1 has a larger macroscopic conductance than O2. In the revised manuscript, the authors performed single-channel recordings to determine why O1 and O2 have different macroscopic conductance. The results show that at voltages where the state O1 predominates, channels exhibited longer open times and overall higher open probability, whereas at more depolarized voltages where occupancy of O2 increases, channels exhibited more flickery gating behavior and decreased open probability. These results are informative but not conclusive because additional details about how experiments were conducted, and group data analysis are missing. Importantly, results showing inhibition of single Δ PASCap channels by a Kv10-specific inhibitor are mentioned but not shown or quantitated - these data are essential to establish that the new O1 conductance indeed represents Kv10 channel activity.

It is shown that conditioning pulses to very negative voltages result in mutant channel currents that are larger and activate more slowly than those elicited at the same voltage but starting from less negative conditioning pulses. In voltage-activated curves, O1 occupancy is shown to be favored by increasingly negative conditioning voltages. This is interpreted as indicating that O1 is primarily accessed from deeply closed states in which voltage sensors are in their most deactivated position. Consistently, a mutation that destabilizes these deactivated states is shown to largely suppress the first component in voltage-activation curves for both Δ PASCap and E600R channels.

The authors then address the role of the hidden O1 state in channel regulation by calmodulation. Stimulating calcium entry into oocytes with ionomycin and thapsigargin, assumed to enhance CaM-dependent modulation, resulted in preferential potentiation of the first component in Δ PASCap and E600R channels. This potentiation was attenuated by including an additional mutation that disfavors deeply closed states. Together, these results are interpreted as an indication that calcium-CaM preferentially stabilizes deeply closed states from which O1 can be readily accessed in mutant channels, thus favoring current activation. In WT channels lacking a conducting O1 state, CaM stabilizes deeply closed states and is therefore inhibitory. It is found that the potentiation of Δ PASCap and E600R by CaM is more strongly attenuated by mutations in the channel that are assumed to disrupt interaction with the C-terminal lobe of CaM than mutations assumed to affect interaction with the N-terminal lobe. These results are intriguing but difficult to interpret in mechanistic terms. The strong effect that calcium-CaM had on the occupancy of the O1 state in the mutants raises the possibility that O1 can be only observed in channels that are constitutively associated with CaM. To address this, a biochemical pull-down assay was carried out to establish that only a small fraction of channels are associated with CaM under baseline conditions. These CaM experiments are potentially very interesting and could have wide physiological relevance. However, the approach utilized to activate CaM is indirect and could result in additional non-specific effects on the oocytes that could affect the results.

Finally, a mathematical model is proposed consisting of two layers involving two activation steps for the voltage sensor, and one conformational change in the cytoplasmic gating ring - completion of both sets of conformational changes is required to access state O2, but accessing state O1 only requires completion of the first voltage-sensor activation step in the four subunits. The model qualitatively reproduces most major findings on the mutants. Although the model used is highly symmetric and appears simple, the mathematical form used for the rate constants in the model adds a layer of complexity to the model that makes mechanistic interpretations difficult. In addition, many transitions that from a mechanistic standpoint should not depend on voltage were assigned a voltage dependence in the model. These limitations diminish the overall usefulness of the model which is prominently presented in the manuscript. The most important mechanistic assumptions in the model are not addressed experimentally, such as the proposition that entry into O1 depends on the opening of the transmembrane pore gate, whereas entry into O2 involves gating ring

transitions - it is unclear why O2 would require further gating ring transitions to conduct ions given that the gating ring can already support permeation by O1 without any additional conformational changes.

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Reviewer #3 (Public Review):

In the present manuscript, Abdelaziz and colleagues interrogate the gating mechanisms of Kv10.1, an important voltage-gated K⁺ channel in cell cycle and cancer physiology. At the molecular level, Kv10.1 is regulated by voltage and Ca-CaM. Structures solved using Cryo-EM for Kv10.1 as well as other members of the KCNH family (Kv11 and Kv12) show channels that do not contain a structured S4-S5 linker imposing therefore a non-domain swapped architecture in the transmembrane region. However, the cytoplasmatic N- and C- terminal domains interact in a domain swapped manner forming a gating ring. The N-terminal domain (PAS domain) of one subunit is located close to the intracellular side of the voltage sensor domain and interacts with the C-terminal domain (CNBHD domain) of the neighbor subunit. Mutations in the intracellular domains has a profound effect in the channel gating. The complex network of interactions between the voltage-sensor and the intracellular domains makes the PAS domain a particularly interesting domain of the channel to study as responsible for the coupling between the voltage sensor domains and the intracellular gating ring.

The coupling between the voltage-sensor domain and the gating ring is not fully understood and the authors aim to shed light into the details of this mechanism. In order to do that, they use well established techniques such as site-directed mutagenesis, electrophysiology, biochemistry and mathematical modeling. In the present work, the authors propose a two open state model that arises from functional experiments after introducing a deletion on the PAS domain (Δ PAS Cap) or a point mutation (E600R) in the CNBHD domain. The authors measure a bi-phasic G-V curve with these mutations and assign each phase as two different open states, one of them not visible on the WT and only unveiled after introducing the mutations. The hypothesis proposed by the authors could change the current paradigm in the current understanding for Kv10.1 and it is quite extraordinary; therefore, it requires extraordinary evidence to support it.

STRENGTHS: The authors use adequate techniques such as electrophysiology and site-directed mutagenesis to address the gating changes introduced by the molecular manipulations. They also use appropriate mathematical modeling to build a Markov model and identify the mechanism behind the gating changes.

WEAKNESSES: The results presented by the authors do not fully support their conclusions since they could have alternative explanations. The authors base their primary hypothesis on the bi-phasic behavior of a calculated G-V curve that do not match the tail behavior, the experimental conditions used in the present manuscript introduce uncertainties, weakening their conclusions and complicating the interpretation of the results. Therefore, their experimental conditions need to be revisited

I have some concerns related to the following points:

(1) Biphasic gating behavior

The authors use the TEVC technique in oocytes extracted surgically from *Xenopus Leavis* frogs. The method is well established and is adequate to address ion channel behavior. The experiments are performed in chloride-based solutions which present a handicap when measuring outward rectifying currents at very depolarizing potentials due to the presence of calcium activated chloride channel expressed endogenously in the oocytes; these channels

will open and rectify chloride intracellularly adding to the outward rectifying traces during the test pulse.

The authors calculate their G-V curves from the test pulse steady-state current instead of using the tail currents. The conductance measurements are normally taken from the 'tail current' because tails are measured at a fix voltage hence maintaining the driving force constant. Calculating the conductance from the traces should not be a problem, however, in the present manuscript, the traces and the tail currents do not agree. The tail traces shown in Fig1E do not show an increasing current amplitude in the voltage range from +50mV to +120mV, they seem to have reached a 'saturation state', suggesting that the traces from the test pulse contain an inward chloride current contamination. In addition, this second component identified by the authors as a second open state appears after +50mV and seems to never saturate. The normalization to the maximum current level during the test pulse, exaggerates this second component on the calculated G-V curve. It's worth noticing that the Δ PASCap mutant experiments on Fig 5 in Mes based solutions do not show that second component on the G-V.

Because these results are the foundation for their two open state hypotheses, I will strongly suggest the authors to repeat all their Chloride-based experiments in Mes-based solutions to eliminate the undesired chloride contribution to the mutants current and clarify the contribution of the mutations to the Kv10.1 gating.

(2) Two step gating mechanism.

The authors interpret the results obtained with the Δ PASCap and the E600R as two step gating mechanisms containing two open states (O1 and O2) and assign them to the voltage sensor movement and gating ring rotation respectively. It is not clear, however how the authors assign the two open states.

The results show how the first component is conserved amongst mutations; however, the second one is not. The authors attribute the second component, hence the second open state to the movement of the gating ring. This scenario seems unlikely since there is a clear voltage-dependence of the second component that will suggest an implication of a voltage-sensing current.

The split channel experiment is interesting but needs more explanation. I assume the authors expressed the 2 parts of the split channel (1-341 and 342-end), however Tomczak et al showed in 2017 how the split presents a constitutively activated function with inward currents that are not visible here, this point needs clarification.

Moreover, the authors assume that the mutations introduced uncover a new open state, however the traces presented for the mutations suggest that other explanations are possible. Other gating mechanisms like inactivation from the closed state, can be introduced by the mutations. The traces presented for Δ PASCap but specially E600R present clear 'hooked tails', a direct indicator of a populations of inactive channels during the test pulse that recover from inactivation upon repolarization (Tristani-Firouzi M, Sanguinetti MC. J Physiol. 1998). The results presented by the authors can be alternatively explained with a change in the equilibrium between the close to inactivated/recovery from inactivation to the open state. Finally, the authors state that they do not detect "cumulative inactivation after repeated depolarization" but that is considering inactivation only from the open state and ignoring the possibility of the existence of close state inactivation or, that like in hERG, that the channel inactivates faster that what it activates (Smith PL, Yellen G. J Gen Physiol. 2002).

(3) Single channel conductance.

The single channels experiments are a great way to assess the different conductance of single channel openings, unfortunately the authors cannot measure accurately different conductances for the two proposed open states. The Markov Model built by the authors, disagrees with their interpretation of the experimental results assigning the exact same

conductance to the two modeled open states. To interpret the mutant data, it is needed to add data with the WT for comparison and in presence of specific blockers.

<https://doi.org/10.7554/eLife.91420.2.sa1>

Author response:

The following is the authors' response to the current reviews.

Gating of Kv10 channels is unique because it involves coupling between non-domain swapped voltage sensing domains, a domain-swapped cytoplasmic ring assembly formed by the N- and C-termini, and the pore domain. Recent structural data suggests that activation of the voltage sensing domain relieves a steric hindrance to pore opening, but the contribution of the cytoplasmic domain to gating is still not well understood. This aspect is of particular importance because proteins like calmodulin interact with the cytoplasmic domain to regulate channel activity. The effects of calmodulin (CaM) in WT and mutant channels with disrupted cytoplasmic gating ring assemblies are contradictory, resulting in inhibition or activation, respectively. The underlying mechanism for these discrepancies is not understood. In the present manuscript, Reham Abdelaziz and collaborators use electrophysiology, biochemistry and mathematical modeling to describe how mutations and deletions that disrupt inter-subunit interactions at the cytoplasmic gating ring assembly affect Kv10.1 channel gating and modulation by CaM. In the revised manuscript, additional information is provided to allow readers to identify within the Kv10.1 channel structure the location of E600R, one of the key channel mutants analyzed in this study. However, the mechanistic role of the cytoplasmic domains that this study focuses on, as well as the location of the Δ PASCap deletion and other perturbations investigated in the study remain difficult to visualize without additional graphical information. This can make it challenging for readers to connect the findings presented in the study with a structural mechanism of channel function.

The authors focused mainly on two structural perturbations that disrupt interactions within the cytoplasmic domain, the E600R mutant and the Δ PASCap deletion. By expressing mutants in oocytes and recording currents using Two Electrode Voltage-Clamp (TEV), it is found that both Δ PASCap and E600R mutants have biphasic conductance-voltage (G-V) relations and exhibit activation and deactivation kinetics with multiple voltage-dependent components. Importantly, the mutant-specific component in the G-V relations is observed at negative voltages where WT channels remain closed. The authors argue that the biphasic behavior in the G-V relations is unlikely to result from two different populations of channels in the oocytes, because they found that the relative amplitude between the two components in the G-V relations was highly reproducible across individual oocytes that otherwise tend to show high variability in expression levels. Instead, the G-V relations for all mutant channels could be well described by an equation that considers two open states O1 and O2, and a transition between them; O1 appeared to be unaffected by any of the structural manipulations tested (i.e. E600R, Δ PASCap, and other deletions) whereas the parameters for O2 and the transition between the two open states were different between constructs. The O1 state is not observed in WT channels and is hypothesized to be associated with voltage sensor activation. O2 represents the open state that is normally observed in WT channels and is speculated to be associated with conformational changes within the cytoplasmic gating ring that follow voltage sensor activation, which could explain why the mutations and deletions disrupting cytoplasmic interactions affect primarily O2.

Severing the covalent link between the voltage sensor and pore reduced O1 occupancy in one of the deletion constructs. Although this observation is consistent with the hypothesis

that voltage-sensor activation drives entry into O1, this result is not conclusive. Structural as well as functional data has established that the coupling of the voltage sensor and pore does not entirely rely on the S4-S5 covalent linker between the sensor and the pore, and thus the severed construct could still retain coupling through other mechanisms, which is consistent with the prominent voltage dependence that is observed. If both states O1 and O2 require voltage sensor activation, it is unclear why the severed construct would affect state O1 primarily, as suggested in the manuscript, as opposed to decreasing occupancy of both open states. In line with this argument, the presence of Mg²⁺ in the extracellular solution affected both O1 and O2. This finding suggests that entry into both O1 and O2 requires voltage-sensor activation because Mg²⁺ ions are known to stabilize the voltage sensor in its most deactivated conformations.

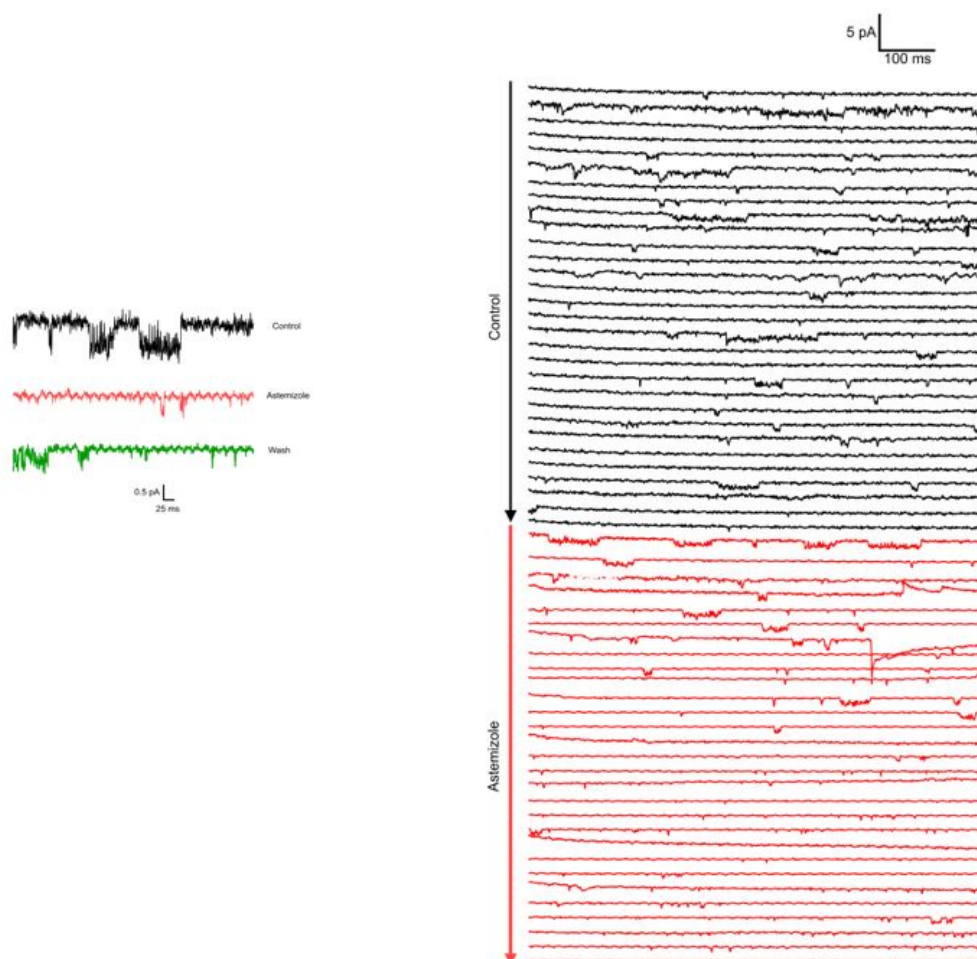
We agree with the reviewer that access to both states requires a conformational change in the voltage sensor. This was stated in our revised article: “In contrast, to enter O2, all subunits must complete both voltage sensor transitions and the collective gating ring transition.” We interpret the two gating steps as sequential; the effective rotation of the intracellular ring would happen only once the sensor is in its fully activated position.

We also agree that the S4-S5 segment cannot be the only interaction mechanism, as we demonstrated in our earlier work (Lörinczi et al., 2015; Tomczak et al., 2017).

Activation towards and closure from O1 is slow, whereas channels close rapidly from O2. A rapid alternating pulse protocol was used to take advantage of the difference in activation and deactivation kinetics between the two open components in the mutants and thus drive an increasing number of channels towards state O1. Currents activated by the alternating protocol reached larger amplitudes than those elicited by a long depolarization to the same voltage. This finding is interpreted as an indication that O1 has a larger macroscopic conductance than O2. In the revised manuscript, the authors performed single-channel recordings to determine why O1 and O2 have different macroscopic conductance. The results show that at voltages where the state O1 predominates, channels exhibited longer open times and overall higher open probability, whereas at more depolarized voltages where occupancy of O2 increases, channels exhibited more flickery gating behavior and decreased open probability. These results are informative but not conclusive because additional details about how experiments were conducted, and group data analysis are missing. Importantly, results showing inhibition of single Δ PASCap channels by a Kv10-specific inhibitor are mentioned but not shown or quantitated - these data are essential to establish that the new O1 conductance indeed represents Kv10 channel activity.

We observed the activity of a channel compatible with Kv10.1 Δ PAS-Cap (long openings at low-moderate potentials, very short flickery activity at strong depolarizations) in 12 patches from oocytes obtained from different frog operations over a period of two and a half months once the experimental conditions could be established. As stated in the text, we did not proceed to generate amplitude histograms because we could not resolve clear single-channel events at strong depolarizations. Astemizole abolished the activity and (remarkably) strongly reduced the noise in traces at strong depolarizations, which we interpret as partially caused by flicker openings.

Author response image 1.



We include two example recordings of Astemizole application (100 μ M) on two different patches. Both recordings are performed at -60 mV (to decrease the likelihood that the channel visits O2) with 100 mM internal and 60 mM external K⁺. In both cases, the traces in Astemizole are presented in red.

It is shown that conditioning pulses to very negative voltages result in mutant channel currents that are larger and activate more slowly than those elicited at the same voltage but starting from less negative conditioning pulses. In voltage-activated curves, O1 occupancy is shown to be favored by increasingly negative conditioning voltages. This is interpreted as indicating that O1 is primarily accessed from deeply closed states in which voltage sensors are in their most deactivated position. Consistently, a mutation that destabilizes these deactivated states is shown to largely suppress the first component in voltage-activation curves for both Δ PASCap and E600R channels.

The authors then address the role of the hidden O1 state in channel regulation by calmodulation. Stimulating calcium entry into oocytes with ionomycin and thapsigargin, assumed to enhance CaM-dependent modulation, resulted in preferential potentiation of the first component in Δ PASCap and E600R channels. This potentiation was attenuated by including an additional mutation that disfavors deeply closed states. Together, these results are interpreted as an indication that calcium-CaM preferentially stabilizes deeply closed states from which O1 can be readily accessed in mutant

channels, thus favoring current activation. In WT channels lacking a conducting O1 state, CaM stabilizes deeply closed states and is therefore inhibitory. It is found that the potentiation of Δ PASCap and E600R by CaM is more strongly attenuated by mutations in the channel that are assumed to disrupt interaction with the C-terminal lobe of CaM than mutations assumed to affect interaction with the N-terminal lobe. These results are intriguing but difficult to interpret in mechanistic terms. The strong effect that calcium-CaM had on the occupancy of the O1 state in the mutants raises the possibility that O1 can be only observed in channels that are constitutively associated with CaM. To address this, a biochemical pull-down assay was carried out to establish that only a small fraction of channels are associated with CaM under baseline conditions. These CaM experiments are potentially very interesting and could have wide physiological relevance. However, the approach utilized to activate CaM is indirect and could result in additional nonspecific effects on the oocytes that could affect the results.

Finally, a mathematical model is proposed consisting of two layers involving two activation steps for the voltage sensor, and one conformational change in the cytoplasmic gating ring - completion of both sets of conformational changes is required to access state O2, but accessing state O1 only requires completion of the first voltage-sensor activation step in the four subunits. The model qualitatively reproduces most major findings on the mutants. Although the model used is highly symmetric and appears simple, the mathematical form used for the rate constants in the model adds a layer of complexity to the model that makes mechanistic interpretations difficult. In addition, many transitions that from a mechanistic standpoint should not depend on voltage were assigned a voltage dependence in the model. These limitations diminish the overall usefulness of the model which is prominently presented in the manuscript. The most important mechanistic assumptions in the model are not addressed experimentally, such as the proposition that entry into O1 depends on the opening of the transmembrane pore gate, whereas entry into O2 involves gating ring transitions - it is unclear why O2 would require further gating ring transitions to conduct ions given that the gating ring can already support permeation by O1 without any additional conformational changes.

In essence, we agree with the reviewer; we already have addressed these points in our revised article:

Regarding the voltage dependence we write “the κ/λ transition could reasonably be expected to be voltage independent because we related it to ring reconfiguration, a process that should occur as a consequence of a prior VSD transition. We have made some attempts to treat this transition as voltage independent but state-specific with upper-layer bias for states on the right and lower-layer bias for states on the left. This is in principle possible, as can already be gleaned from the similar voltage ranges of the left-right transition (α/β) and the κ/λ transition. However, this approach leads to a much larger number of free, less well constrained kinetic parameters and drastically complicated the parameter search.” As you can see, we also formulated a strategy to free the model of the potentially spurious voltage dependence and (in bold here) explained why we did not follow this route in this study.

Regarding the need for gating ring transitions after O1, we wrote, “Thus, the underlying gating events can be separated into two steps: The first gating step involves only the voltage sensor without engaging the ring and leads to a pre-open state, which is non-conducting in the WT but conducting in our mutants. The second gating event operates at higher depolarizations, involves a change in the ring, and leads to an open state both in WT and in the mutants.”

We interpret your statements such that you expect the conducting state to remain available once O1 is reached. However, the experimental evidence speaks against that the pore

availability remains regardless of the further gating steps beyond O1. The description of model construction is informative here: "... we could exclude many possible [sites at which O1 connects to closed states] because the attachment site must be sufficiently far away from the conventional open state [O2]. Otherwise, the transition from "O1 preferred" to "O2 preferred" via a few closed intermediate states is very gradual and never produces the biphasic GV curves [that we observed]."

In other words, voltage-dependent gating steps beyond the state that offers access to O1 appear to close the pore, after it was open. That might occur because only then (for states in which at least one voltage sensor exceeded the intermediate position) the ring is fixed in a particular state until all sensors completed activation. In the WT, closing the pore in deactivated states might rely on an interaction that is absent in the mutant because, at least in HERG: "the interaction between the PAS domain and the C-terminus is more stable in closed than in open Kv11.1 (HERG) channels, and a single chain antibody binding to the interface between PAS domain and CNBHD can access its epitope in open but not in closed channels, strongly supporting a change in conformation of the ring during gating"

Reviewer #3 (Public Review):

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The hypothesis proposed by the authors could change the current paradigm in the current understanding for Kv10.1 and it is quite extraordinary; therefore, it requires extraordinary evidence to support it.

STRENGTHS: *The authors use adequate techniques such as electrophysiology and sitedirected mutagenesis to address the gating changes introduced by the molecular manipulations. They also use appropriate mathematical modeling to build a Markov model and identify the mechanism behind the gating changes.*

WEAKNESSES: *The results presented by the authors do not fully support their conclusions since they could have alternative explanations. The authors base their primary hypothesis on the bi-phasic behavior of a calculated G-V curve that do not match the tail*

behavior, the experimental conditions used in the present manuscript introduce uncertainties, weakening their conclusions and complicating the interpretation of the results. Therefore, their experimental conditions need to be revisited.

We respectfully disagree. We think that your suggestions for alternative explanations are addressed in the current version of the article. We will rebut them once more below, but we feel the need to point out that our arguments are already laid out in the revised article.

I have some concerns related to the following points:

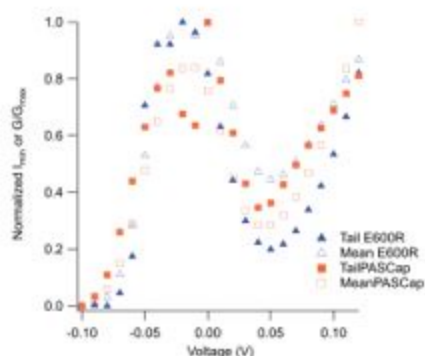
(1) Biphasic gating behavior

*The authors use the TEVC technique in oocytes extracted surgically from *Xenopus Leavis* frogs. The method is well established and is adequate to address ion channel behavior. The experiments are performed in chloride-based solutions which present a handicap when measuring outward rectifying currents at very depolarizing potentials due to the presence of calcium activated chloride channel expressed endogenously in the oocytes; these channels will open and rectify chloride intracellularly adding to the outward rectifying traces during the test pulse. The authors calculate their G-V curves from the test pulse steady-state current instead of using the tail currents. The conductance measurements are normally taken from the 'tail current' because tails are measured at a fix voltage hence maintaining the driving force constant.*

We respectfully disagree. In contrast to other channels, like HERG, a common practice for Kv10 is not to use tail currents. It is long known that in this channel, tail currents and test-pulse steady-state currents can appear to be at odds because the channels deactivate extremely rapidly, at the border of temporal resolution of the measurements and with intricate waveforms. This complicates the estimation of the instantaneous tail current. Therefore, the outward current is commonly used to estimate conductance (Terlau et al., 1996; Schönherr et al., 1999; Schönherr et al., 2002; Whicher and MacKinnon, 2019), while the latter authors also use the extreme of the tail for some mutants.

Due to their activation at very negative voltage, the reversal potential in our mutants can be measured directly; we are, therefore, more confident with this approach. Nevertheless, we have determined the initial tail current in some experiments. The behavior of these is very similar to the average that we present in Figure 1. The biphasic behavior is unequivocally present.

Author response image 2.



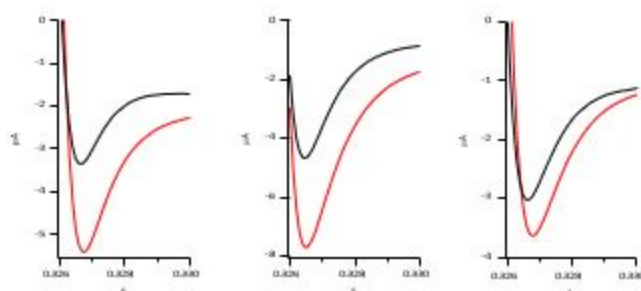
Calculating the conductance from the traces should not be a problem, however, in the present manuscript, the traces and the tail currents do not agree.

The referee's observation is perfectly in line with the long-standing experience of several labs working with KV10: tail current amplitudes in KV10 appear to be out of proportion for the WT open state (O2). Importantly, this is due to the rapid closure, which is not present in O1. As a consequence, the initial amplitude of tail currents from O1 are easier to estimate correctly, and they are much more obvious in the graphs. Taken together, these differences between O1 and O2 explain the misconception the reviewer describes next.

The tail traces shown in Fig1E do not show an increasing current amplitude in the voltage range from +50mV to +120mV, they seem to have reached a 'saturation state', suggesting that the traces from the test pulse contain an inward chloride current contamination.

As stated in the text and indicated in Author response image 3, the tail currents in Figure 1E increase in amplitude between +50 and +120 mV, as can be seen in the examples below from different experiments (+50 is presented in black, +120 in red). As stated above, the increase is not as evident as in traces from other mutants because the predominance of O2 also implies a much faster deactivation.

Author response image 3.



We are aware that Ca^{2+} -activated Cl^- currents can represent a problem when interpreting electrophysiological data in oocytes. In fact, we show in Supplement 1 to Figure 8 that this can be the case during the Ca^{2+} -CaM experiments, where the increase in Ca^{2+} would certainly augment Cl^- contribution to the outward current. This is why we performed these experiments in Cl^- -free solutions. As we show in Figure 8, the biphasic behavior was also present in those experiments.

Importantly, Cl^- free bath solutions would not correct contamination during the tail, since this would correspond to Cl^- exiting the oocyte. Yet, if there would be contamination of the outward currents by Cl^- , one would expect it to increase with larger depolarizations as the typical Ca^{2+} -activated Cl^- current in oocytes does. As the reviewer states, this does not seem to be the case.

In addition, this second component identified by the authors as a second open state appears after +50mV and seems to never saturate. The normalization to the maximum current level during the test pulse, exaggerates this second component on the calculated G-V curve.

We agree that this second component continues to increase; the reviewer brought this up in the first review, and we have already addressed this in our reply and in the discussion of the revised version: “This flicker block might also offer an explanation for a feature of the mutant channels, that is not explained in the current model version: the continued increase in current amplitude, hundreds of milliseconds into a strong depolarization (Supp. 4 to Fig. 9). If the relative stability of O2 and C2 continued to change throughout depolarization, such a current creep-up could be reproduced. However, this would require either the introduction of further layers of $O_n \leftrightarrow C_n$ states, or a non-Markovian modification of the model’s time evolution.” With non-Markovian, we mean a Langevin-type diffusive process.

It’s worth noticing that the Δ PASCap mutant experiments on Fig 5 in Mes based solutions do not show that second component on the G-V.

For the readers of this conversation, we would like to clarify that the reviewer likely refers to experiments shown in Fig. 5 of the initial submission but shown in Fig. 6 of the revised version (“Hyperpolarization promotes access to a large conductance, slowly activating open state.” Fig. 5 deals with single channels). We agree that these data look different, but this is because the voltage protocols are completely different (compare Fig. 6A (fixed test pulse, varied prepulse) and Fig. 2A (varied test pulse, fixed pre-pulse). Therefore, no biphasic behavior is expected.

Because these results are the foundation for their two open state hypotheses, I will strongly suggest the authors to repeat all their Chloride-based experiments in Mes-based solutions to eliminate the undesired chloride contribution to the mutants current and clarify the contribution of the mutations to the Kv10.1 gating.

In summary, we respectfully disagree with all concerns raised in point (1). Our detailed arguments rebutting them are given above, but there is a more high-level concern about this entire exchange: the referee casts doubt on observations that are not new. Several labs have reported for a group of mutant KCNH channels: non-monotonic voltage dependence of activation (see, e.g., Fig. 6D in Zhao et al., 2017), multi-phasic tail currents (see e.g. Fig. 4A in Whicher and MacKinnon, 2019, in CHO cells where Cl- contamination is not a concern), and activation by high $[Ca^{2+}]_i$ (Lörinczi et al., 2016). Our study replicates those observations and hypothesizes that the existence of an additional conducting state can alone explain all previously unexplained observations. We highlight the potency of this hypothesis with a Markov model that qualitatively reproduces all phenomena. We not only factually disagree with the individual points raised, but we also think that they don’t touch on the core of our contribution

(2) Two step gating mechanism.

The authors interpret the results obtained with the Δ PASCap and the E600R as two step gating mechanisms containing two open states (O1 and O2) and assign them to the voltage sensor movement and gating ring rotation respectively. It is not clear, however how the authors assign the two open states.

The results show how the first component is conserved amongst mutations; however, the second one is not. The authors attribute the second component, hence the second open state to the movement of the gating ring. This scenario seems unlikely since there is a clear voltage dependence of the second component that will suggest an implication of a voltage-sensing current.

We do not suggest that the gating ring motion is not voltage dependent. We would like to point out that voltage dependence can be conveyed by voltage sensor coupling to the ring;

this is the widely accepted theory of how the ring can be involved. Should the reviewer mean it in a narrow sense, that the model should be constructed such that all voltage-dependent steps occur before and independently of ring reconfiguration and that only then an additional step that reflects the (voltage-independent) reconfiguration solely, we would like to point the reviewer to the article, where we write: “the κ/λ transition could reasonably be expected to be voltage independent because we related it to ring reconfiguration, a process that should occur as a consequence of a prior VSD transition. We have made some attempts to treat this transition as voltage independent but state-specific with upper-layer bias for states on the right and lower-layer bias for states on the left. This is in principle possible, as can already be gleaned from the similar voltage ranges of the left-right transition (α/β) and the $\kappa L/\lambda$ transition. However, this approach leads to a much larger number of free, less well constrained kinetic parameters and drastically complicated the parameter search.” As you can see, we also formulated a strategy to free the model from the potentially spurious voltage dependence and (in bold here) explained why we did not follow this route in this study.

The split channel experiment is interesting but needs more explanation. I assume the authors expressed the 2 parts of the split channel (1-341 and 342-end), however Tomczak et al showed in 2017 how the split presents a constitutively activated function with inward currents that are not visible here, this point needs clarification.

As stated in the panel heading, the figure legend, and the main text, we did not use 1-341 and 342-end as done in Tomczak et al. Instead, “we compared the behavior of $\Delta 2-10$ and $\Delta 210.L341Split$,”. Evidently, the additional deletion (2-10) causes a shift in activation that explains the difference you point out. However, as we do not compare $L341Split$ and $\Delta 210.L341Split$ but $\Delta 2-10$ and $\Delta 2-10.L341Split$, our conclusion remains that “As predicted, compared to $\Delta 2-10$, $\Delta 2-10.L341Split$ showed a significant reduction in the first component of the biphasic GV (Fig. 2C, D).” Remarkably, the behavior of the $\Delta 3-9 L341Split$ described in Whicher and MacKinnon, 2019 (Figure 5) matches that of our $\Delta 2-10 L341Split$, which we think reinforces our case.

Moreover, the authors assume that the mutations introduced uncover a new open state, however the traces presented for the mutations suggest that other explanations are possible. Other gating mechanisms like inactivation from the closed state, can be introduced by the mutations. The traces presented for $\Delta PASCap$ but specially $E600R$ present clear 'hooked tails', a direct indicator of a populations of inactive channels during the test pulse that recover from inactivation upon repolarization (Tristani-Firouzi M, Sanguinetti MC. J Physiol. 1998).

There is a possibility that we are debating nomenclature here. In response to the suggestion that all our observations could be explained by inactivation, we attempted a disambiguation of terms in the reply and the article. As the argument is brought up again without reference to our clarification attempts, we will try to be more explicit here:

If, starting from deeply deactivated states, an open state is reached first, and then, following further activation steps, closed states are reached, this might be termed “inactivation”. In such a reading, our model features many inactivated states. The shortest version of such a model is C-O-I. It is for instance used by Raman and Bean (2001; DOI: 10.1016/S00063495(01)76052-3) to explain NaV gating in Purkinje neurons. If “inactivation” is meant in the sense that a gating transition exists, which is orthogonal to an activation/deactivation axis, and that after this orthogonal transition, an open state cannot be reached anymore, then all of the upper floor in our model is inactivated with respect to the open state O1. Finally, the state C2 is an inactivated state to O2. In this view, “inactivation” explains the observed phenomena.

However, we must disagree if the referee means that a parsimonious explanation exists in which a single conducting state is the only source for all observed currents.

There is a high-level reason: we found a single assumption that explains three different phenomena, while the inactivation hypothesis with one conducting state cannot explain one of them (the increase of the first component under raised CaM). But there is also a low-level reason: the tails in Tristani-Firouzi and Sanguinetti 1998 are fundamentally different from what we report herein in that they lack a third component. Thus, those tails are consistent with recovery from inactivation through a single open state, while a three-component tail is not. In the framework of a Markov model, the time constants of transitions from and to a given state (say O2), cannot change unless the voltage changes. During the tail current, the voltage does not change, yet we observe:

i) a rapid decrease with a time constant of at most a few milliseconds (Fig 9 S2, 1-> 2), ii) a slow increase in current, peaking after approximately 25 milliseconds and iii) a relaxation to zero current with a time constant of >50 ms.

According to the reviewer's suggestion, these processes on three timescales should all be explained by depopulating and repopulating the same open state while all rates are constant. There might well be a complicated multi-level state diagram with a single open state with different variants, like (open and open inactivated) that could produce triphasic tails with these properties if the system had not reached a steady state distribution at the end of the test pulse. It cannot, however, achieve it from an equilibrated system, and certainly, it cannot at the same time produce "biphasic activation" and "activation by CaM".

The results presented by the authors can be alternatively explained with a change in the equilibrium between the close to inactivated/recovery from inactivation to the open state.

Again, we disagree. The model construction explains in detail that the transition from the first to the second phase is not gradual. Shifting equilibria cannot reproduce this. We have extensively tested that idea and can exclude this possibility.

Finally, the authors state that they do not detect "cumulative inactivation after repeated depolarization" but that is considering inactivation only from the open state and ignoring the possibility of the existence of close state inactivation or, that like in hERG, that the channel inactivates faster than what it activates (Smith PL, Yellen G. J Gen Physiol. 2002).

We respectfully disagree. We explicitly model an open state that inactivates faster (O2->C2) than it activates. Once more, this is stated in the revised article, which we point to for details. Again, this alternative mechanism does not have the potential to explain all three effects. As discussed above about the chloride contamination concerns, this inactivation hypothesis was mentioned in the first review round and, therefore, addressed in our reply and the revised article. We also explained that "inactivation" has no specific meaning in Markov models. In the absence of O1, all transitions towards the lower layer are effectively "inactivation from closed states", because they make access to the only remaining open state less likely". But this is semantics. What is relevant is that no network of states around a single open state can reproduce the three effects in a more parsimonious way than the assumption of the second open state does.

(3) Single channel conductance.

The single channels experiments are a great way to assess the different conductance of single channel openings, unfortunately the authors cannot measure accurately different conductances for the two proposed open states. The Markov Model built by the authors,

disagrees with their interpretation of the experimental results assigning the exact same conductance to the two modeled open states. To interpret the mutant data, it is needed to add data with the WT for comparison and in presence of specific blockers.

We respectfully disagree. As previously shown, the conductance of the flickering wild-type open state is very difficult to resolve. Our recordings do not show that the two states have different single-channel conductances, and therefore the model assumes identical singlechannel conductance.

The important point is that the single-channel recordings clearly show two different gating modes associated with the voltage ranges in which we predict the two open states. One has a smaller macroscopic current due to rapid flickering (aka “inactivation”). These recordings are another proof of the existence of two open states because the two gating modes occur. Wild-type data can be found in Bauer and Schwarz, (2001, doi:10.1007/s00232-001-0031-3) or Pardo et al., (1998, doi:10.1083/jcb.143.3.767) for comparison.

We appreciate the effort editors and reviewers invested in assessing the revised manuscript. Yet, we think that the demanded revision of experimental conditions and quantification methods contradicts the commonly accepted practice for KV10 channels. Some of the reviewer comments are skeptical about the biphasic behavior, which is an established and replicated finding for many mutants and by many researchers. The alternative explanations for these disbelieved findings are either “semantics” or cannot quantitatively explain the measurements. Therefore, only the demand for more explanations and unprecedented resolution in singlechannel recordings remains. We share these sentiments.

——— The following is the authors’ response to the original reviews.

(1) The authors must show that the second open state is not just an artifact of endogenous activity but represents the activity of the same EAG channels. I suggest that the authors repeat these experiments in Mes-based solutions.

(2) Along the same lines, it is necessary to show that these currents can be blocked using known EAG channel blockers such as astemizole. Ultimately, it will be important to demonstrate using single-channel analysis that these do represent two distinct open states separated by a closed state.

We have addressed these concerns using several approaches. The most substantial change is the addition of single-channel recordings on Δ PASCap. In those experiments, we could provide evidence of the two types of events in the same patch, and the presence of an outward current at -60 mV, 50 mV below the equilibrium potential for chloride. The channels were never detected in uninjected oocytes, and Astemizole silenced the activity in patches containing multiple channels. These observations, together with the maintenance of the biphasic behavior that we interpret as evidence of the presence of O1 in methanesulfonate-based solutions, strongly suggest that both O1 and O2 obey the expression of KV10.1 mutants.

(3) Currents should be measured by increasing the pulse lengths as needed in order to obtain the true steady-state G-V curves.

We agree that the endpoint of activation is ill-defined in the cases where a steady-state is not reached. This does indeed hamper quantitative statements about the *relative amplitude* of the two components. However, while the overall shape does change, its position (*voltage dependence*) would not be affected by this shortcoming. The data, therefore, supports the claim of the “existence of mutant-specific O1 and its equal voltage dependence across mutants.”

(4) A more clear and thorough description should be provided for how the observations with the mutant channels apply to the behavior of WT channels. How exactly does state O1 relate to WT behavior, and how exactly do the parameters of the mathematical model differ between WT and mutants? How can this be interpreted at a structural level? What could be the structural mechanism through which Δ PASCap and E600R enable conduction through O1? It seems contradictory that O1 would be associated exclusively with voltage-sensor activation and not gating ring transitions, and yet the mutations that enable cation access through O1 localize at the gating ring - this needs to be better clarified.

We have undertaken a thorough rewriting of all sections to clarify the structural correlates that may explain the behavior of the mutants. In brief, we propose that when all four voltage sensors move towards the extracellular side, the intracellular ring maintains the permeation path closed until it rotates. If the ring is altered, this “lock” is incompetent, and permeation can be detected (page 34). By fixing the position of the ring, calmodulin would preclude permeation in the WT and promote the population of O1 in the mutants.

(5) Rather than the t80% risetime, exponential fits should be performed to assess the kinetics of activation.

We agree that the assessment of kinetics by a t80% is not ideal. We originally refrained from exponential fits because they introduce other issues when used for processes that are not truly exponential (as is the case here). We had planned to perform exponential fits in this revised version, but because the activation process is not exponential, the time constants we could provide would not be accurate, and the result would remain qualitative as it is now. In the experiments where we did perform the fits (Fig. 3), the values obtained support the statement made.

(6) It is argued based on the G-V relations in Figure 2A that none of the mutations or deletions introduced have a major effect on state O1 properties, but rather affect state O2. However, the occupancy of state O2 is undetermined because activation curves do not reach saturation. It would be interesting to explore the fitting parameters on Fig.2B further to test whether the data on Fig 2A can indeed only be described by fits in which the parameters for O1 remain unchanged between constructs.

We agree that the absolute occupancy of O2 cannot be properly determined if a steady state is not reached. This is, however, a feature of the channel. During very long depolarizations in WT, the current visually appears to reach a plateau, but a closer look reveals that the current keeps increasing after very long depolarizations (up to 10 seconds; see, e.g., Fig. 1B in Garg et al., 2013, Mol Pharmacol 83, 805-813. DOI: 10.1124/mol.112.084384). Interestingly, although the model presented here does not account for this behavior, we propose changes in the model that could. “If the relative stability of O2 and C2 continued to change throughout the depolarization such a current creep-up could be reproduced. However, this would require either the introduction of further layers of On $\leftrightarrow$ Cn states or a non-Markovian modification of the model’s evolution.” Page 34.

(7) The authors interpret the results obtained with the mutants DPASCAP and E600R - tested before by Lorinczi et al. 2016, to disrupt the interactions between the PASCap and cNBHD domains- as a two-step gating mechanism with two open states. All the results obtained with the E600R mutant and DPASCap could also be explained by inactivation/recovery from inactivation behavior and a change in the equilibrium between the closed states closed/inactivated states and open states. Moreover, the small tails between +90 to +120 mV suggest channels accumulate in an inactive state (Fig 1E).

It is not convincing that the two open-state model is the mechanism underlying the mutant's behavior.

We respectfully disagree with the notion that a single open state can provide a plausible explanation for "All the results obtained with the E600R mutant and DPASCap". We think that our new single channel results settle the question, but even without this direct evidence, a quantitative assessment of the triphasic tail currents all but excludes the possibility of a single open state. We agree that it is, in principle, possible to obtain some form of a multiphasic tail with a single open state using the scheme suggested in this comment: at the end of the test pulse, a large fraction of the channels must be accumulated in inactive states, and a few are in the open state. The hyperpolarization to -100mV then induces a rapid depopulation of the open state, followed by slower replenishments from the inactive state. Exactly this process occurs in our model, when C2 empties through O2 (Supp. 5 to Fig 9, E600R model variant). However, this alone is highly unlikely to quantitatively explain the measured tail currents, because of the drastically different time scales of the initial current decay (submillisecond to at most a few milliseconds lifetime) and the much slower transient increase in current (several tens of milliseconds) and the final decay with time constants of >100 ms (see for instance data in Fig. 1 E for E600R +50 to +120mV test pulse). To sustain the substantial magnitude of slowly decaying current by slow replenishment of an open state with a lifetime of 1 ms requires vast amounts of inactivated channels. A rough estimation based on the current integral of the initial decay and the current integral of the slowly decaying current suggests that at the end of the test pulse, the ratio inactivated/open channels would have to be 500 to 1500 for this mechanism to quantitatively explain the observed tail currents. To put this in perspective: This would suggest that without inactivation all the expressed channels in an oocyte would provide 6 mA current during the +100 mV test pulse. While theoretically possible, we consider this a less likely explanation than a second open state.

(8) Different models should be evaluated to establish whether the results in Figure 4 can also be explained by a model in which states O1 and O2 have the same conductance. It would be desirable if the conductance of both states were experimentally determined - noise analysis could be applied to estimate the conductance of both states.

In the modified model, O1 and O2 have the same single-channel conductance. The small conductance combined with the fast flickering did not allow an accurate determination, but we can state that there is no evidence that the single-channel conductance of the states is different.

(9) Although not included, it looks like the model predicts some "conventional inactivation" This can be appreciated in Fig 8, and in the traces at -60mV. Interestingly, the traces obtained in the absence of Cl⁻ also undergo slow inactivation, or 'conventional inactivation' as referred to by the authors. Please revise the following statement "Conventional inactivation was never detected in any mutants after repeated or prolonged depolarization. In the absence of inactivation, the pre-pulse dependent current increase at +40 mV could be related to changes in the relative occupancy of the open states".

We have carefully edited the manuscript to address this concern. The use of the term inactivation admittedly represents a challenge. We agree that the state that results from the flickering block (C2) could be defined as "inactivated" because it is preceded by an open state. Yet, in that case, the intermediate states that the channel travels between O1 and O2 would also be *sensu stricto* "inactivated", but only in the mutants. We have made this clear in page 17.

Recommendations for improving the writing and presentation.

(1) Methods section: Please state the reversal potential calculated for the solution used. It looks like the authors used an Instantaneous I-V curve method to calculate the reversal potential; if that's correct, please show the I-V and the traces together with the protocol used.

We have provided the calculated reversal potentials for excised patches. We cannot predict the reversal potential in whole oocytes because we have no control over the intracellular solution. The reversal potential was determined in the mutants through the current at the end of the stimulus because the mutants produced measurable inward currents. The differences in reversal potential were not significant among mutants.

Pulse protocols have been added to the figures.

(2) Figure 1 suggestion: Combine the two panels in panel D and move the F panel up so the figure gets aligned in the lower end.

Thank you, this has been done.

(3) Please clarify the rationale for using the E600R-specific mutant. I assume it is based on the Lorinzi et al. 2016 effect and how this is similar to the DPASCap phenotype, or is it due to the impact of this mutation in the interactions between the N-term and the CNBHD?

We have explained the rationale for the use of E600R explicitly on page 6.

(4) Fig S1A is not present in the current version of the manuscript. Include a cartoon as well as a structural figure clearly depicting the perturbations introduced by E600R, ΔPASCap, and the other deletions that are tested. Additional structural information supporting the discussion would also be helpful to establish clearer mechanistic links between the experimental observations described here and the observed conformational changes between states in Kv10 channel structures.

We have corrected this omission, thank you for pointing it out.

(5) It would be informative to see the traces corresponding to the I-V shown in Fig 7 A and B at the same indicated time points (0, 60, 150, and 300s). Did the authors monitor the Ca²⁺ signal rise after the I&T treatment to see if it coincides with the peak in the 60s?

In Figure 7 (now Figure 8) we used voltage ramps instead of discrete I-V protocols because of the long time required for recording the latter. This is stated on page 19. Ca²⁺ was monitored through Cl⁻ current after ionomycin/thapsigargin. The duration of the Ca²⁺ increase was reproducible among oocytes and in good agreement with the changes observed in the biphasic behavior of the mutants (Supplement 1 to Figure 8).

(6) Fig 4. Please state in the legend what the different color traces correspond to in E600R and DPASCap. Is there a reason to change the interpulse on DPASCap to -20mV and not allow this mutant to close? Please state. How do the authors decide the 10 ms interval for the experiments in Fig 2?

Thank you for pointing this out, we have added the description. We have explained why we use a different protocol for ΔPASCap and the reason for using 10 ms interval (we believe the referee means Figure 4) on page 12.

(7) Fig. 5. Since the pre-pulse is supposed to be 5s, but the time scale doesn't correspond with a pre-pulse of 5 s before the test pulse to +40mV. Has the pre-pulse been trimmed for representation purposes? If so, please state.

The pre-pulse was 5s, but as the reviewer correctly supposed, the trace is trimmed to keep the +40 mV stimulus visible. This has now been clearly stated in the legend.

(8) The mutant L322H is located within the S4 helix according to the Kv10.1 structure (PDB 5K7L), not in the 'S3-S4 linker'; please correct.

This has been done, thank you.

The introduction of this mutant should also shift the voltage dependence toward more hyperpolarizing potentials (around 30mV, according to Schoenherr et al. 1999). It looks like that shift is present within the first component of the G-V. Still, since the max amplitude from the second component could be contaminated by endogenous Cl⁻ currents, this effect is minimized. Repeating these experiments in the no Cl⁻ solutions will help clarify this point and see the effect of the DPASCap and E600R in the background of a mutation that accelerates the transitions between the closed states (see Major comment 1). Did the authors record L322H alone for control purposes?

We have decided not to measure L322H alone or repeat the measurements in Cl⁻-free solutions because we do not see a way to use the quantitative assessment of the voltage dependence of L322H and the L322H-variants of the eag domain mutants. Like in our answer to main point 3, we base our arguments not on the precise voltage dependence of the second component but on the shape of the G-V curves instead, specifically the consistent appearance of the first component and the local conductance minimum between the first and second components. After the introduction of L322H the first component is essentially absent.

We think that the measurements of the L322H mutants cannot be interpreted as a hyperpolarizing shift in the first component. The peak of the first conductance component occurs around -20 mV in ΔPASCap and E600R (Fig. 7 C, D). After a -30mV shift, in L322H+DPASCap and L322H+E600R, this first peak would still be detected within the voltage range in our experiments, but it is not. A contamination of the second component would have little impact on this observation, which is why we refrain from the suggested measurements.

(9) The authors differentiate between an O1 vs. O2 state with different conductances, and maybe I missed it, but there's no quantitative distinction between the components; how are they different?

Please see the response to the main comments 1 and 2. This has been addressed in singlechannel recordings.

(10) Please state the voltage protocols, holding voltages, and the solutions (K⁺ concentration and Cl⁻ presence/absence) used for the experiments presented in the legends on the figures. Hence, it's easier to interpret the experiments presented.

Thank you, this has been done.

(11) The authors state on page 7 that "with further depolarizations, the conductance initially declined to rise again in response to strong depolarizations. This finding matches the changes in amplitude of the tail currents, which, therefore, probably reflect a true

change in conductance" However, the tails in the strong voltage range (+50 to +120 mV) for the E600R mutant argue against this result. Please review.

The increase in the amplitude of the tail current is also present in E600R, but the relative increase is smaller. We have decided against rescaling these traces because the Figure is already rather complex. We indicated this fact with a smaller arrow and clarified it in the text (page 8).

(12) The authors mention that the threshold of activation for the WT is around -20mV; however, the foot of the G-V is more around -30 or -40mV. Please revise.

Thank you. We have done this.

(13) The authors state on page 9 that the 'second component occurs at progressively more depolarized potentials for increasingly larger N-terminal deletions" However E600R mutant that conserves the N-terminal intact has a shift as pronounced as the DPASCap and larger than the D2-10. How do the authors interpret this result?

We have corrected this statement in page 10 : "...the second component occurs at progressively more depolarized potentials for increasingly larger N-terminal deletions and when the structure of the ring is altered through disruption of the interaction between N- and C-termini (E600R)".

(14) The equation defined to fit the G-Vs, can also be used to describe the WT currents. If the O1 is conserved and present in the WT, this equation should also fit the WT data properly. The 1-W component shown could also be interpreted as an inactivating component that, in the WT, shifts the voltage-dependence of activation towards depolarizing potentials and is not visible. Still, the mutants do show it as if the transition from closed-inactivated states is controlled by interactions in the gating ring, and disturbing them does affect the transitions to the open state.

Out of the two open states in the mutant, O2 is the one that shares properties with the WT (e.g. it is inaccessible during Ca²⁺-CaM binding) while O1 is the open state with the voltage dependence that is conserved across the mutants. We, therefore, believe that this question is based on a mix-up of the two open states. We appreciate the core of the question: does the pattern in the mutants' G-V curves find a continuation in the WT channel?

Firstly, the component that is conserved among mutants does not lead to current in the WT because the corresponding open state (O1) is not observed in WT. However, the gating event represented by this component should also occur in WT and –given its apparent insensitivity to eag domain mutations– this gating step should occur in WT with the same voltage dependence as in all the mutants. This means that this first component sets a hard boundary for the most hyperpolarized G-V curve we can expect in the WT, based on our mutant measurements. Secondly, the second component shows a regular progression across mutants: The more intact the eag domain is, the more hyperpolarized the V_{half} values of transition term (1-W) and O2 activation. In Δ2-10, the transition term already almost coincides with O1 activation (estimated V_{half} values of -33.57 and -33.47 mV). A further shift of (1-W) in the WT is implausible because, if O1 activation is coupled to the earliest VSD displacement, the transition should not occur before O1 activation. Still, the second component might shift to more hyperpolarized values in the WT, depending on the impact of amino acids 2 to 10 on the second VSD transition.

In summary, in WT the G-V should not be more hyperpolarized than the first component of the mutants, and the (1-W)-component probably corresponds to the Δ2-10 (1-W)-component. In WT the second component should be no more *depolarized* than the second component of

$\Delta 2-10$. The WT G-V (Fig.1B) meets all these predictions derived from the pattern in the mutant GV: When we use Eq. 4 to fit the WT G-V with $A1=0$ (O1 is not present in WT) and the parameters of the transition term (1-W) fixed to the values attained in $\Delta 2-10$, we obtain a fit for the O2 component with $V_{half}=+21mV$. This value nicely falls into the succession of V_{half} values for Δe_{ag} , $\Delta PASC_{ap}$, and $\Delta 2-10$ (+103mV,+80mV,+52mV) and, at the same time, it is not more hyperpolarized than the conserved first component ($V_{half} -34mV$). Our measurements therefore support that the O2 component in the mutants corresponds to the single open state in the WT.

(15) Page 15, the authors state that 'The changes in amplitude and kinetics in response to rising intracellular Ca^{2+} support our hypothesis that Ca-CaM stabilized O1, possibly by driving the channels to deep closed states (Fig 5 and 6)' (pg 15). This statement seems contradictory; I can't quite follow the rationale since Ca^{2+} potentiates the current (Fig 7), and the addition of the L322H mutant in Fig 7 makes the shift of the first component to negative potentials visible.

Please check the rationale for this section.

We have explained this more explicitly in the discussion (page 32). “Because access to O1 occurs from deep closed states, this could be explained by an increased occupancy of such deactivated states in response to CaM binding. This appears to be the case since CaM induces a biphasic behavior in the mutant channels that show reduced access to deep closed states; thus, L322H mutants behave like the parental variants in the presence of Ca^{2+} -CaM. This implies a mechanistic explanation for the effect of Ca^{2+} -CaM on WT since favoring entry into deep closed states would result in a decrease in current amplitude in the absence of (a permeable) O1”.

Also, Figs 5 and 6 seem miscited here.

Thank you, we have corrected this.

(16) For Figure 5, it would be helpful if each of the current traces corresponding to a particular voltage had a different color. That way, it will be easier to see how the initial holding voltage modulates current.

We have considered this suggestion, and we agree that it would make it easier to follow. Yet, since we have identified the mutants with different colors, it would be inconsistent if we used another color palette for this Figure. Supplement 3 to Figure 9 shows the differences in a clearer way.

(17) Add zero-current levels to all current traces.

We have done this.

(18) The mathematical model should be described better. Particularly, the states from which O1 can be accessed should be described more clearly, as well as whether the model considers any direct connectivity between states O1 and O2. The origin of the voltage-dependence for transitions that do not involve voltage-sensor movements should be discussed. Also, it separation of κ into κ_{p-l} and κ_{p-r} should be described.

We have extensively rewritten the description of the mathematical model to address these concerns.

(19) Page 4, "reveals a pre-open state in which the transmembrane regions of the channel are compatible with ion permeation, but is still a nonconducting state". Also, page 27, "renders a hydrophobic constriction wider than 8 Å, enough to allow K⁺ flow, but still corresponds to a non-conducting state". These sentences are confusing - how can the regions be compatible with ion permeation, and still not be conducting? Is cation conductance precluded by a change in the filter, or elsewhere? How is it established that it represents a non-conducting state?

We have rephrased to clarify this apparent inconsistency. Page 4: "(...) in which the transmembrane regions of the channel are compatible with ion permeation (the permeation path is dilated, like in open states) but the intracellular gate is still in the same conformation as in closed states (Zhang et al., 2023)." Page 31: "The presence of an intact intracellular ring would preclude ionic flow in the WT, and its alteration would explain the permeability of this state in the mutants."

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